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Drone Discover

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by

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أشهد أنّي اطّلعت على قوانين الكلية الخاصّة بتقييم الطّلبة الموجود بالكتاب التقديمي للجامعة.

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Dedication

We dedicate this work to our beloved families, dearest friends, the Computers and Information Technology College, Taif University and to our supervisor **dr. Hesham Al-humyani** who supported us to do a great work and complete this project at its best form to gain the bachelor's degree.

Also, this project is dedicated to the group members for their effort and cooperation researching it. And we want to thank our families one more time for their unwavering support both morally and financially during the time of developing this project, they were a beacon of wisdom that always reminded us that even the biggest tasks can be accomplished if it is done in pieces. And so many thanks to our committee for their guidance and great effort of evaluating our project.

In the end, we hope this project or idea helps as many people as possible.

Abstract

The Kingdom suffers from slums and visual distortion, and municipalities are trying hard to solve this problem. Because of the expansion and proliferation of these regions, it is difficult for human staff to resolve them in record time. And the most important causes of destruction of the urban landscape of cities and neighborhoods caused are Citizens' lack of awareness. To address this problem, we designed a drone to identify visual distortions such as general hygiene, drilling barriers, street excavations, lampposts, rundown pavements, and construction residues by locating and filming. Also, organize neighborhoods by YOLO algorithm. This drone is characterized by the organization of random neighborhood's, accuracy in identifying the offence and its location and scanning a greater distance in a shorter time to accomplish tasks.

Keywords: Drone - Development of slums - Visual distortion - Random

ملخص مقترح البحث (عربي)

تعاني المملكة من المناطق العشوائية وما تحويه من تشوه بصري حيث تحاول البلديات جاهده لحل هذه المشكلة لكن بسبب توسع هذه المناطق وانتشارها يصعب على الكوادر البشرية حلها في وقت قياسي. ومن أهم أسباب تدمير المشهد العمراني للمدن والأحياء هو قلة وعي المواطنين.

لمعالجه هذه المشكلة قمنا بتصميم طائرة دون طيار لتحديد التشوهات البصرية مثل النظافة العامة، وحواجز الحفر، وحفر الشوارع، وأعمدة الإنارة، والأرصفة المتهدمة، ومخلفات البناء عن طريق تحديد المواقع والتصوير. أيضًا، قم بتنظيم الأحياء باستخدام خوارزمية YOLO. تتميز هذه الطائرة بتنظيم حي عشوائي ودقة في تحديد المخالفة وموقعها ومسحها لمسافة أكبر في وقت أقصر لإنجاز المهام.

الكلمات المفتاحية: طائرة بدون طيار، تطوير العشوائيات، التشوه البصري، عشوائي.

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List of Acronyms (Abbreviations) and Symbol

IOT	Internet of Things
UAV	Unmanned Aerial Vehicles
CNN	Convolutional neural networks
RPA	remotely piloted aircraft
VTOL	vertical takeoff and landing vehicle
NR-IQA	no-reference image quality assessment
3D	Three-dimensional space
2D	Two-dimensional space
GCS	Ground Control Stations
UAS	Unmanned Aircraft System
RPAS	Remotely Piloted Aircraft System
RC	Remote Control
DARPA’s	Defense Advanced Research Projects Agency
MAVs	Mars Ascent Vehicle
HALE	High-Altitude Long Endurance
ASD	Available Sight Distance
AI	Artificial Intelligence
GPS	Global Positioning System
COCO	Common Objects in Context

YOLO	You Only Look Once
ResNet	Residual Network
IMU	Inertial Measurement Unit
PID	Proportional Integral Derivative
EKF	Extended Kalman Filter
GPIO	General-Purpose Inputs and Outputs
SoC	System-on-Chip
BLE	Bluetooth Low Energy
CSI	Camera interface
DSI	Display interface
PDB	Power distribution board
PWM	Pulse Width Modulation
IGBT	Insulated Gate Bipolar Transistor
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
ESC	Electronics speed controller

Chapter 1: Introduction

1 Introduction

A Drone has the ability to perform many tasks that humans cannot, such as monitoring high temperatures and high altitudes in many industries. Therefore, there are many features where we can reduce the visual deformation inherent in slums and use a fire detection mechanism and rainwater harvesting areas because these random areas have poor discharge. We can also reduce street encroachment and organise neighbourhoods. It is important to solve these problems and reduce uncivilised violations.

1.1 Overview and Problem Background

This project helps to capture images to reduce many problems within slums such as visual distortion, water collection due to poor drainage by taking images where the Drone is connected to my system and uploading the problem image and location of the system. It also helps to solve the problem of unregulated slums by identifying abuses on the streets by reducing the use of human resources and saving time and effort.

Using laser technology to determine whether the street is an outlet or not and to determine the street's width

1.2 Problem Statement

The problem of the destruction of the urban landscape of cities and neighborhoods such as: the problem of visual mutilation and some hazards such as difficulty of drainage in neighborhoods causing the accumulation of water and the problem of unregulated slums (street encroachments).

1.3 Objectives

This project aims to achieve the following objectives:

- Organize randoms using 3D laser scan. to identify domestic-based violations and abuses.
- Monitor and report all visual distortions to remove them.
- Fire detection and reporting.
- Detect the accumulation of rainwater as a result of poor drainag**e**

1.4 Contributions/Significance of the Project

The project provides several tasks, first it scans the random area, and then the algorithms determine the area that contains optical distortion, and also monitor the accumulation of water. Thus, the photos are sent to the system with the location of the violation

1.5 Scope

The project targets people interested in the field of environment, municipalities, students who are researchers in the same field to develop their projects, the Civil Aviation , and Civil Defense Authority to monitor natural disasters

1.6 Engineering Standards

- The International System of Units (SI) used in our project:

Voltage (Volt)
Current (mA)
Frequency (MHz)
Temperature (Degree Celsius C°)
Angle (Degree °)
Length (mm)
weight (g)
width (mm)

Table 1.6.1 System of Units

To meet this set of outcomes, our circuit diagrams of Sensors and raspberry pi that follows the Standard Reference Designations for Electrical and Electronic parts and Equipment.

Sensor&processors	Specification
Barometer:MS5611	to measure atmospheric pressure
Accel/Gyro	ICM-42688-P BMI055
ESC hw30a motor driver	Is a electronic circuit that manages and regulates an electric motor's speed
Mag	IST8310
GPS	Locating the drone
IMU(inertial measurement unit)	**Gyroscope:** Calculates the amount of change in the angle of the drone's fuselage **Accelerometer:** to calculate the acceleration of the drone's fuselage **Magnetometer:** To determine the direction of the fuselage of the drone
FMU Processor: STM32H743	32 Bit Arm® Cortex®-M7, 480MHz, 2MB memory, 1MB SRAM
IO Processor: STM32F103	32 Bit Arm® Cortex®-M3, 72MHz, 64KB SRAM

Table 1.6.2 Measuring Unit

1.7 Realistic Constraints

•**Economic**: The Kingdom of Saudi Arabia seeks to get rid of visual distortion and slums by achieving the "Quality of Life" Program, which is one of the initiatives of Vision 2030, for various reasons, the most important of which is the improvement of the civilized landscape and the upgrading of the services provided.

The slum areas often lack the necessary services such as live drainage, water, electricity, health and education, so that slums impede the economy of the state and limit its development. The potential cost is 1000 riyals, the cost of the components is 873 riyals, and the project is marketed at about 2000 riyals in the beginning depending on the size of the project and its expansion in the labor market.

•**Environment**: It facilitates the detection of water pools and reduces epidemics in slum areas. It will also provide a good solution to reduce slums and reduce the risks of water pools.

•**Sustainability**: There are no limits to development, but when there is any error in the system, it is updated and developed

•**Manufacturability**: We need factories in the Kingdom of Saudi Arabia to manufacture drones so that they become a supply and support for upcoming projects

•**Ethical**: The ethical issues we face in this project are Noises, Espionage, and Privacy. These issues are included under the ethical issues and can be resolved using appropriate solutions. This is a very sensitive point because it is ethical and, in most cases, the ethical problem comes first

•**Health and Safety**: The advantages that our project provides on human health are safety from diseases resulting from water gatherings such as dengue fever and bacterial diseases.

•**Social**: The social effects resulted from the low level of education, through the organization of slums through artificial intelligence

1.8 Development Time Frame and Cost

The schedule is a significant part of the project. It defines what you intend to do and when you plan to do it. You should consider how long each activity will take, which activities must precede others, and how much overlap is possible or desirable. The schedule identifies tasks to be performed, milestones to be met, and the estimated number of hours for each task.

Table 1.8.1 Semester 1 Schedule

Week	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**H**	**%**
Task												
Project proposal	12	8	6	6	6	6					44	20.3%
Study of current research work			12	12	4	4					32	14.8%
Problem formulation			12	12	4	4					32	14.8%
Model design							12	12	8	4	36	16.6%
CPI Report						16	12	8	8	4	48	22.3%
Presentation						4	4	4	4	8	24	11.2%
Hours	12	8	30	30	14	34	28	24	20	16	216	100%

Table 1.8.2 Semester 2 Schedule

Week	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**H**	**%**
Task												
System implementation	14	14	16	16	14						74	34.9%
System integration and testing				14	6	6	10				36	17%
Final report					12	12	6	6	8		44	20.8%
Project poster							8	8	4		20	9.4%
Presentation							8	8	12	10	38	17.9%
Hours	14	14	16	30	32	18	32	22	24	10	212	100%

Components Bill of Materials:

Components	Price in USD	Price in SAR
Raspberry Pi ٣ B+ and 4 B	159.82 USD	600 SAR
S500 V2 Full Kit-Pixhawk **Pixhawk 6C Flight Controller** **PM02 V3-12S Power Module** **M8N GPS Module** **SiK Telemetry Radio V3 433/915MHz** **S500 V2 Frame Kit** **Motors - Holybro 2216 KV920 Motor (4 pcs)** **ESCs - BLHeli S ESC 20A (4 pcs)** **1045 Propellers (4 pcs)**	532.72 USD	2000 SAR
Raspberry Pi HQ Camera	88.70 USD	333 SAR
Lipo Battery	106.54 USD	400 SAR
Wifi Adapter	64 USD	240 SAR
RC Transmitter and receiver	69 USD	259
Total	1020.78 USD	3832 SAR

Table 1.8.3 Cost of Component

Chapter 2: Literature Review

2.1) Overview of related technology

Aerial vehicles have been used for industrial purposes since the 19th century. Balloons have been used to take images since 1860 for distant sensing applications [1]. A breast-mounted airborne camera has been carried by pigeons since 1903 for photography [2]. Aerial torpedoes, sometimes known as the forerunners of drones, were created around the start of World War I [3,4]. Unmanned aerial vehicle research and development has drawn more of the author's attention from academic and business sectors worldwide [5,6].

According to [7], anyone may now buy a drone (a term with a variety of connotations), and it can be difficult to discern the purposes for which people may or may not utilize the drone. Since many urban areas now have so-called "no flight zones" to prevent interfering with specified operations, people need special licenses to operate drones. However, even if a drone is utilized within the bounds set by the law, it might not be secure. It is often possible to use remotely controlled vehicles. The identification of the sorts and intentions of such items has since required investigation.

In [8], drones were initially used in the military and were only made available to consumers afterwards. The autonomy and response time of the drones are also taken into account while making enhancements to their use in surveillance. Commercial enterprise sectors have begun to choose drones as time- and cost-efficient instruments. By establishing waypoints that the drone can follow, Unmanned Aerial Vehicle UAV has also been used in [9] to move little goods and parcels without expending a lot of computational energy.

The latency in recognizing and tracking moving objects has been studied as a concern with drones' capacity to recognize and track items in [10]. This idea has been crucial in the case of using such technology in an industry's monitoring system. This idea might also be used to detect when an object is out of place and deliver an input indicating the need to move it back into position. Giving the drone the ability to carry out recognition operations without the assistance of the control center was the most practical way to handle this problem.

Many phrases can refer to the same thing in [11], remotely piloted aircraft (RPA): "an aircraft without a pilot on board" and controlled either externally or by a computer. These words have been expanded to include everything connected to using a UAV, including controllers, Ground Control Stations (GCS), and launching pads, as well as UA System (UAS) or RPA System (RPAS). Any type of unmanned aerial vehicle—including those that cannot be controlled, such missiles—is referred to as a UAV. The term "drone" refers to either autonomous or remotely controlled (RC) aircrafts and was a subclass of UAV. Finally, as implied by its name, the RPA exclusively covers radio-controlled aircraft.

Drone, (UAV), has been utilized and selected as the key phrase in [12] for a number of reasons. First of all, it is compatible with contemporary devices that may be utilized with both RC and autonomous schemes simultaneously. Second, it has been in use since the 1930s, when the military first used it.

Numerous approaches have been tested in [13] to enhance actual aircraft with vertical takeoff and landing capabilities. First, in 1928, Nikola Tesla proposed the idea of a vertical takeoff and landing vehicle (VTOL). Single engines with thrust vectoring have been employed in sophisticated VTOL aircraft. Thrust vectoring shows how an engine could be used to control both vertical and horizontal flight by the aircraft by sending thrust in different directions.

One of the most well-known and successful fixed-wing singles-engine VTOL aircraft in [14] was the Harrier Jump Jet. The use of UAVs has become more commonplace in the twenty-first century. Many of these, particularly the quadcopter variety, have VTOL capability. While researching the specifications of DARPA's UAV Forge competition for big and small UAVs, which was posted around the time we started our project, we were also intrigued. As part of the ARES program, Lockheed Martin's Skunk Works collaborated closely with Piasecki Aircraft to develop the next generation of dynamic vertical takeoff and landing (VTOL) transport systems.

UAVs are presently mostly employed for numerous reasons. A drone for agriculture can ascertain the local climate's temperature. The market is driving a number of variables, including the expansion of the agricultural industry, farm machinery, and government efforts [15].

In [16], we were interested in assessing applications for the commercial, government, and industrial sectors. If small drones are widely accessible, new markets and applications will also develop. Reviewing pipelines or actually looking into dangerous areas like an emergency site at an atomic power plant are examples of potential new markets in business and modern applications. Additionally, it appears that small drones could be useful for harvest appraisal or disaster relief.

In [17], unmanned aircraft system (UAS) has been mainly comprised of the (UAV), the ground station (or remote station), the payload, system control and any launching/landing systems.

Similar components to those present in manned aircraft have been established for UAS in [18], with the exception that the manned crew has been substituted by onboard electrical or computer systems. The system can be greatly reduced in size or replaced by other elements that help the UAS function now that people are not inside the air vehicle. As a result, UAS can employ parts and do tasks that a conventionally manned aircraft currently cannot, such as operating entirely on battery power. More intelligent UAVs are able to fly themselves and plan their own flight routes, which is highly helpful in scenarios like video graphing.

Drones are used for both civilian and military purposes throughout a variety of different fields in [19], not just in video graphics. Drones have been widely utilized for surveying in civil applications, such as in agriculture, construction, and conservation.

According to [20], they can also be utilized for security and monitoring, as demonstrated by police departments, fisheries, surveillance, etc. Additionally, have been employed in both inland and coastal search and rescue operations.

However, in [21], the most common uses are for leisure, such as aerial photography, hobbyists, and most recently, drone racing. Reconnaissance and surveillance, as well as airstrikes, are two of the most well-known applications of UAVs in the military. Generally speaking, the use of UAV in military tasks is included in either of these domains, but even so, the range of possible applications is wide, and as a result, drones are now widely used in the military.

According to [22], there are numerous different classifications of aerial drones or UAVs based on their size, weight, function, and method of flight. These many classes will be covered in this part, along with a description of how they work and the sectors in which they are useful. The drone's weight determines the first classification.

According to [23], heavy drones weigh between 200 kg and 2000 kg, whereas extremely heavy drones weigh more than 2000 kg. A Light/Mini drone weighs between 5 kg and 50 kg, whereas a medium drone weighs between 50 kg and 200 kg. Micro air vehicles (MAVs), which are defined as anything weighing less than 5 kg and are the most prevalent drones used for recreational purposes, will be used in this research. The UAVs can also be categorized by their endurance and altitude, which is typically for military uses, rather than by their weight. A

High-Altitude Long Endurance (HALE) drone flies for more than 24 hours at an altitude above 15000 m. A close-range UAV has a range of up to 100 km, whereas a tactical UAV has a range between 100 and 300 km. The MAV and UAV, which have substantially lower ranges, are below this. MAVs / UAVs are the only kind of drones that will work with this design, as was already mentioned.

Road visibility is defined in [24] as the distance or clarity at which road users can perceive the infrastructure of a road. Poor road visibility may make it difficult for drivers to see possible dangers in time, which can lead to traffic accidents. Although some environmental factors, such as light conditions and weather, may affect road visibility, this study gives special attention to the visibility related to sight obstacles, which serve as the main contributors to blind areas and relevant collisions in urban scenes.

In [25], stresses evaluating urban road visibility to vehicle drivers due to the more complicated road environment in urban settings and the higher severity of traffic incidents involving cars. Following the design policy, available sight distance (ASD) and sight triangles (ST) have been employed for many years to give drivers on the road a clear view of the surrounding environment.

There are numerous problems in [26] that ASD and ST are unable to address. Obstacles like plants and information boards may cause poor visibility or blind spots at certain places. Even worse, if drivers are unable to observe the opposing and unyielding road users, a collision may happen. There is also a video of a collision at a non-intersectional unsignalized road entrance. Such driving accidents on city streets have been widely documented in China during the past few years.

However, it has been acknowledged in [27] that neither ASD nor ST adequately capture road visibility. Only the longitudinal road visibility along the vehicle route is taken into account by ASD, and the ST clearance is only evaluated at intersections.

According to [28], agricultural modernization boosts capital income and productivity. Farmers want to use mechanization equipment by embracing new current technology. a smaller, more affordable, and more efficient drone.

In [29], states that this multifunctional drone can be used in agriculture to spray pesticides and fertilizers as well as to sanitize roadways, hallways, and open places. It becomes much more valuable to purchase when it is implemented and made versatile because it serves multiple purposes. According to the most recent data and advancements, forest fires are the main cause of problems in forests. They disrupt the entire flora and fauna, which causes an imbalance in the ecology, biodiversity, and overall environment of the region. Naturally occurring factors such as lightning, low humidity, and high atmospheric temperatures are the primary causes and reasons for forest fires.

According to [30], the no-reference image quality assessment (NR-IQA) technique is frequently used in multimedia content applications because it is more adaptable and ideal for real-time applications. Due of this factor, NR-IQA is a popular subject for academic research. Additionally, NR-IQA is separated into two categories, including NR-IQA procedures that are created for a particular kind of distortion (i.e., Gaussian blur). The other category consists of NR-IQA techniques, which are intended for all applications and do not concentrate on a particular kind of noise or distortion in an image. Instead, they concentrate on evaluating image quality on a broad scale.

Based on the source [31], it describes the primary theoretical ideas behind convolutional neural networks (CNN). CNNs are a result of research on the visual cortex of the brain. On some challenging visual tasks, CNNs have been able to function at a level beyond human capability. Because the visual cortex's neurons have a tiny local receptive field, they only respond to visual inputs that are present in a small portion of the visual field. While some neurons only respond to images of horizontal lines, others only respond to images of lines in other orientations. Additionally, scientists observed that certain neurons have bigger receptive fields and respond to more complicated patterns that combine the simpler patterns.

2.2) Summary of related works

According to the review of the literature, AI technology is improving daily and reducing human mistake and effort. Drone technology is being used in a variety of sectors, including agricultural, aerial surveillance, photography, package delivery, and military. However, there has not been significant advancement in UAVs' use of artificial intelligence technologies.

Drones are an ideal alternative for surveying and mapping since they outperform traditional remote sensing technologies in terms of power consumption, danger to human life, convenience of data gathering, hovering, and ultra-high spatial resolution. Studies confirm the pioneer's findings by highlighting the value and distinction of drones in the civic, logistical, agricultural, and defense sectors.

Drones in Agriculture: Precision farming's main objective is to use the appropriate quantity of input at the right time and location to produce superior results. Data gathering, variability mapping of agricultural areas, data analysis, making farming management decisions based on the outcomes of the analysis, and lastly controlled application such as pesticide spraying and fertilizer application are common techniques of precision farming. The practice of remote sensing utilizing conventional satellite and aerial platforms has been enthusiastically embraced by agriculture.

Drones in Forestry, fisheries and wildlife protection: Drones perform a variety of activities, such as measuring canopy height, following forest wildlife, assisting in forest management, detecting and controlling forest fires, surveying forests, and simply mapping canopy gaps, bridging the gap between satellite imagery's limited availability and cloud cover.

Drones in defense: UAVs were first developed to carry out military duties including spying, intelligence, reconnaissance, and target identification; subsequently, they were used in civil and logistical applications.

Drones for civil applications: All business sectors, from electrical corporations to the railroad industry, are attracted by drones. Drones are becoming more and more popular among electrical businesses as a safer alternative to perilous climbs and power disruptions while inspecting high-tension lines. Drones have been used by railroad firms to monitor and examine track flaws in places with limited access.

2.3) Comparison of related works

Almost all economic sectors, including precision agriculture, logistics, and infrastructure, are developing and using drone technology. Future technologies focused on improving payload, endurance, and human-UAV interaction while also establishing clear norms and regulations to ensure the safe and secure use of UAVs. Additionally, the incorporation of artificial intelligence into drone technology will give the aircraft autonomy over its commanders. At the very least, a competent UAV program must automate the generation of 2D and 3D models and image georectification.

The idea of our project to compare it with previous research drone scans slums, and through laser technology, streets are surveyed for the process of organizing, in addition to taking a picture of visual distortion of all kinds through CNN technology, and fires and water collection are also detected. The model organizes randoms using 3D laser scan. It differs from all previous research because of the combination of all these properties

Chapter 3: System Analysis and Design

3.1) System Model

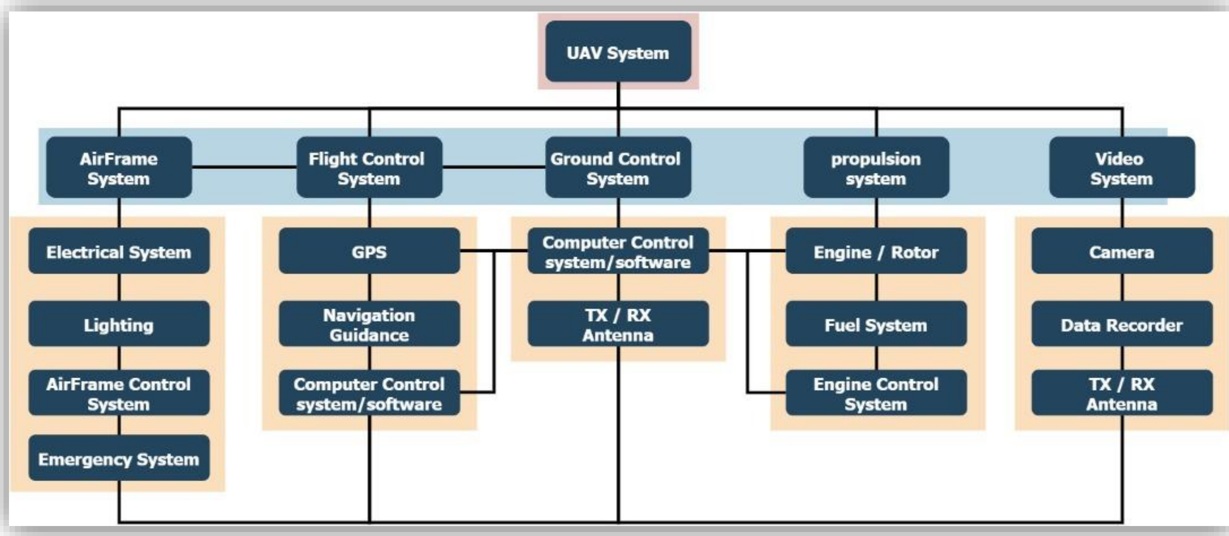


Figure 3.1.1: System Model

We have the UAV System (Unmanned Aerial Vehicles) has:

1- Airframe System represent the main look and contains 4 sections Electrical System with lighting indicators, airframe control system and of course the emergency system which will trigger in a dangerous situation.

2- Flight control system that has a GPS unit with navigation guidance and computer control system and software.

3- Ground Control System which include his own computer control system and software with transmitter and receiver antenna.

4- Propulsion system that contains engines or rotors with fuel system in some cases it's batteries and engine control system.

5- Video system to monitor everything the drone sees which include camera device, recorder, transmitter, and receiver antenna.

3.2) Requirement Specification

3.2.1)- Functional and Non-Functional Requirements

Functional Requirements:

1- GPS receiver
2- flight controller.
3- Cameras.
4- Microcontroller.
5- The UAV must be entirely autonomous.
6- Manual override must be available at all times during flight.
7- The UAV must be able to adjust its flight to avoid obstacles.
8- The UAV shall generate its own waypoints to capture images of entire user defined area and combine all of the images acquired into one high resolution image.
9- The UAV shall alert the user that the survey is complete via municipal system .

Non-Functional Requirements:

1- Security: Does the software comply with any security standards? Does the software collect, store, or send any sensitive user data and how does it protect against any threats.
2- Ease of Use/Usability: The degree of how easy the system should be to be learned or utilized. It illustrates the user experience of the system.
3- Capacity: What is the storage specification of your system?
4- Maintainability: Generally describes the amount of time required to resolve or fix certain key components, changes to be implemented to boost performance.

3.2.2)- Hardware Specifications

List of components:

	
Fig 3.2.2.1: Raspberry pi 4 model B	**Fig 3.2.2.2:** Raspberry pi 3 model B+
	
Fig 3.2.2.3: PDB: Power distribution board	**Fig 3.2.2.4:** WIFI Adapter
	
Fig 3.2.2.5: Flight Controller	**Fig 3.2.2.6:** brushless motor

Fig 3.2.2.7: ESC hw30a motor drive

Fig 3.2.2.8: Raspberry pi HQ Camera

Fig 3.2.2.9: GPS M8N module

Fig 3.2.2.10: Battery

Fig 3.2.2.11: Quadcopter frame

Fig 3.2.2.12:RC transmitter and receiver

Fig 3.2.2.13: PM02 V3-12S Power Module

Fig 3.2.2.14: Radio telemetry (antenna)

- **Components Specification:**

1) Raspberry Pi 3 Model B+:

Figure 3.2.2.1.1: Raspberry Pi 3 Model B+

The Raspberry Pi is a tiny computer the size of a credit card that runs Linux on an ARM processor. The Raspberry Pi 3 Model B+ is the product in question. It contains four USB ports, an Ethernet port, HDMI output, audio output, Bluetooth 4.2, Bluetooth Low Energy (BLE), dual-band WiFi, and 0.1″-spaced pins that allow access to general-purpose inputs and outputs (GPIO). An operating system-containing microSD card is necessary for the Raspberry Pi.

A better model of the Raspberry Pi 3 Model B is the Raspberry Pi 3 Model B+. It is based on the BCM2837B0 system-on-chip (SoC), which has a potent VideoCore IV GPU and a 1.4 GHz quad-core ARMv8 64bit CPU. Together with Microsoft Windows 10 IoT Core, the Raspberry Pi can run the whole spectrum of ARM GNU/Linux distributions, including Snappy Ubuntu Core, Raspbian, Fedora, and Arch Linux.

In addition to having a greater CPU clock speed (1.4 GHz vs. 1.2 GHz), the Raspberry Pi 3 Model B+ also includes dual-band WiFi and enhanced Ethernet throughput. With a Power over Ethernet HAT, it also supports Power over Ethernet.

Features

- 1.4 GHz quad-core BCM2837B0 ARMv8 64bit CPU
- 1 GB RAM
- VideoCore IV 3D graphics core
- Ethernet port
- dual-band (2.4 GHz and 5 GHz) IEEE 802.11.b/g/n/ac wireless LAN (WiFi)
- Bluetooth 4.2
- Bluetooth Low Energy (BLE)
- Four USB ports
- Full-size HDMI output
- Four-pole 3.5 mm jack with audio output and composite video output
- 40-pin GPIO header with 0.1″-spaced male pins that are compatible with our 2×20 stackable female headers and the female ends of our premium jumper wires.
- Camera interface (CSI)
- Display interface (DSI)
- Micro SD card slot

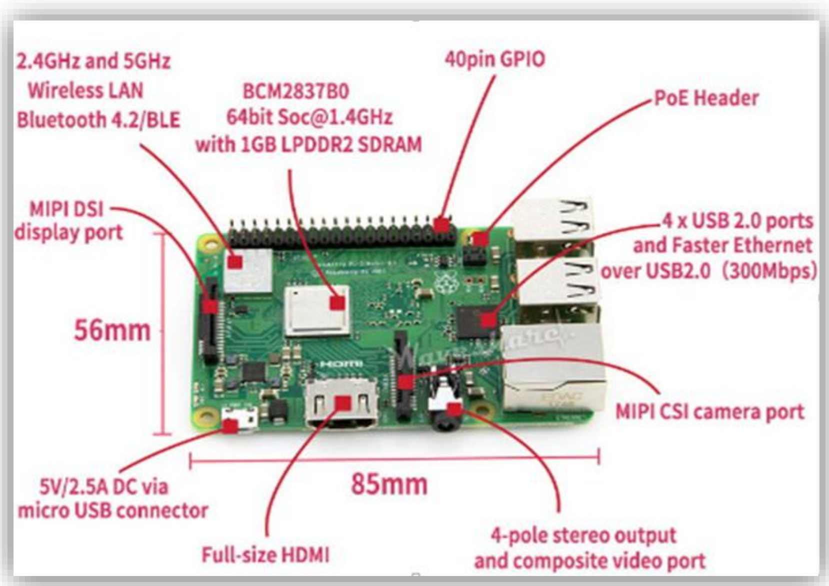


Figure 3.2.2.1.2: Feature of raspberry Pi 3 Model B+

2) Raspberry pi 4 model B:

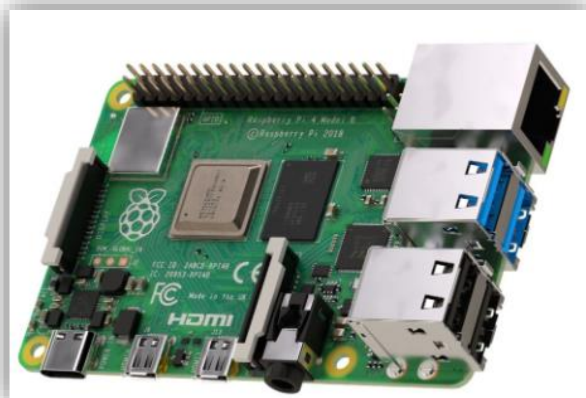


Figure 3.2.2.2.1: Raspberry Pi 4 Model B

The Raspberry Pi is a single board computer which can do a lot of the functions that can be done by your pc, like in games, word process, spreadsheet, and HD videos playback. It is founded in UK by the Raspberry Pi foundation. It's been publicly available from 2012, with the goal of creating a low-cost Academic microcomputer for students. Moreover, it was created with the primary goal of encouraging school-aged pupils to learn, explore, and innovate. It is a small and inexpensive computer. The majority of Raspberry Pi computers are found in cellphones. Mobile computer technologies have grown rapidly in the twentieth century, with the mobile industries driving a large portion of this expansion. ARM technology was used in 98 percent of mobile phones.

There are two models of the Raspberry Pi: model A and model B. The USB port is the main difference between the two models. The Model A board uses less power and does not have an Ethernet connection, while, Model B board does and is made in China. The Raspberry Pi contains a collection of open-source techniques, like multimedia web abilities and communication. The Raspberry Pi Foundation produced the pc module in 2014 to encourage their use, that packs a model B Raspberry Pi board into a module to be used as a component of embedded systems. [32]

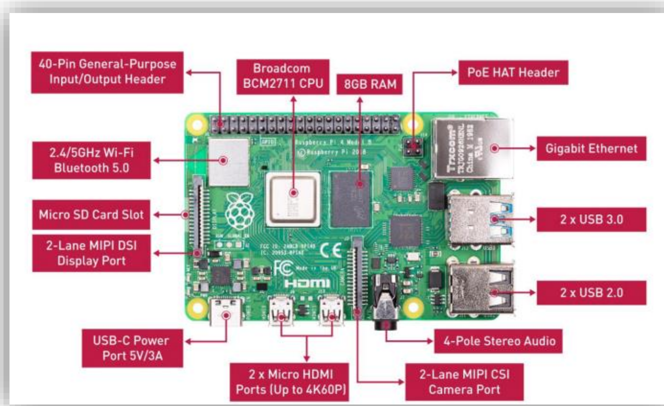


Figure 3.2.2.2.2: Feature of Raspberry Pi 4 Model B

3) PDB: Power distribution board:

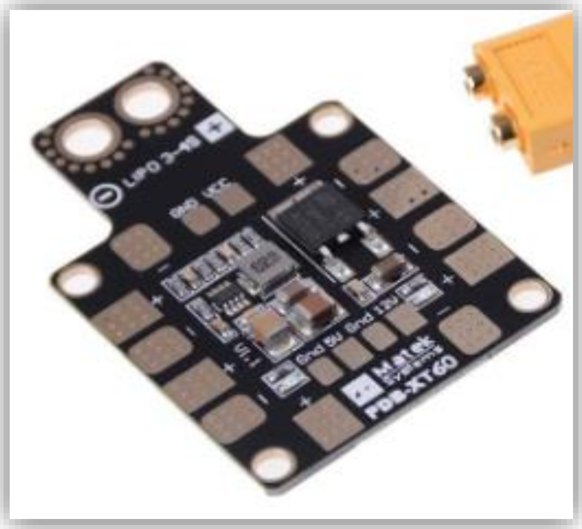


Figure 3.2.2.3: Power distribution board

Is also known as PDB, is an overlooked component simply because it is outdated. Electronic components have come a long way in the past decade. Back in the day, PDB’s were an essential component in the building of a drone. But is it still a viable option? Let’s find out.

Electricity from the battery travels through PDBs and into the drone's esc. Because to technological advancements, PDBs now provide supply power to extra peripherals including Quadcopter Flight Controller, FPV Cameras, and FPV Video Transmitters. Recently, some FCs have built-in PDBs, but because of their size limitations and limited capacity, they can't adequately regulate the voltage spikes caused by the quads' high current demands.

Some individuals, still favor power distribution boards since they make the flight controllers less stressed and typically have greater noise-filtering capabilities. In addition, replacing a single ESC is less expensive than replacing a 4-in-1 ESC.

4) WIFI Adapter

Figure 3.2.2.4: Wi-Fi adapters

A wireless adapter is a hardware device that is generally attached to a computer or other workstation device to allow it to connect to a wireless system. Before the advent of consumer devices with built-in Wi-Fi connectivity, devices required the use of wireless adapters to connect to a network.

Wireless adapters are also known as Wi-Fi adapters

Wireless adapters often come in USB stick form, which must be plugged into a USB port of a computer or device

5) Flight Controller:

Figure 3.2.2.5: Flight Controller

Inside the Pixhawk® 6C, you can find an STMicroelectronics® based STM32H743, paired with sensor technology from Bosch® & InvenSense®, giving you flexibility and reliability for controlling any autonomous vehicle, suitable for both academic and commercial applications.

The Pixhawk® 6C's H7 microcontroller contain the Arm® Cortex®-M7 core running up to 480 MHz, has 2MB flash memory and 1MB RAM. Thanks to the updated processing power, developers can be more productive and efficient with their development work, allowing for complex algorithms and models.

The Pixhawk 6C includes high-performance, low-noise IMUs on board, designed to be cost effective while having IMU redundancy. A vibration isolation System to filter out high-frequency vibration and reduce noise to ensure accurate readings, allowing vehicles to reach better overall flight performances.

UART Mapping¶

- SERIAL0 -> USB
- SERIAL1 -> UART7 (Telem1) RTS/CTS pins
- SERIAL2 -> UART5 (Telem2) RTS/CTS pins
- SERIAL3 -> USART1 (GPS1)
- SERIAL4 -> UART8 (GPS2)
- SERIAL5 -> USART2 (Telem3) RTS/CTS pins
- SERIAL6 -> USART3 (USER)
- SERIAL7 -> USB (can be used for SLCAN with protocol change)

RC Input¶

- The RCIN pin, which by default is mapped to a timer input, can be used for all ArduPilot supported receiver protocols, except CRSF/ELRS and SRXL2 which require a true UART connection. However, FPort, when connected in this manner, will only provide RC without telemetry.

PWM Output¶

- The Pixhawk6C supports up to 16 PWM outputs. All 16 outputs support all normal PWM output formats. All FMU outputs (marked "FMU PWM Output") also support DShot.

Battery Monitoring¶

- The board has 2 dedicated power monitor ports with a 6 pin connector. The Pixhawk6C uses analog power monitors on these ports.

6) Brushless motor:

Figure 3.2.2.6.1: Brushless Motor

While brush motors and brushless DC motors both work using the same magnetic attraction and repulsion principles, they are built slightly differently. The magnetic field of the stator is rotated by using electronic commutation as opposed to a mechanical commutator and brushes. Electronics with active control are needed for this.

In a brushless motor, the stator has windings, and the rotor is attached by permanent magnets. As demonstrated above, brushless motors can have the rotor on the inside of the windings or the outside (referred to as a "out runner" motor).

The phrase "number of phases" refers to the total number of windings in a brushless motor. Although different numbers of phases can be used to build brushless motors, three phase brushless motors are the most frequent. Small cooling fans that only require one or two phases are an exception.

A brushless motor's three windings are connected either in a "delta" or a "star" pattern. The drive method and waveform are the same in both cases, and there are three wires connected to the motor.

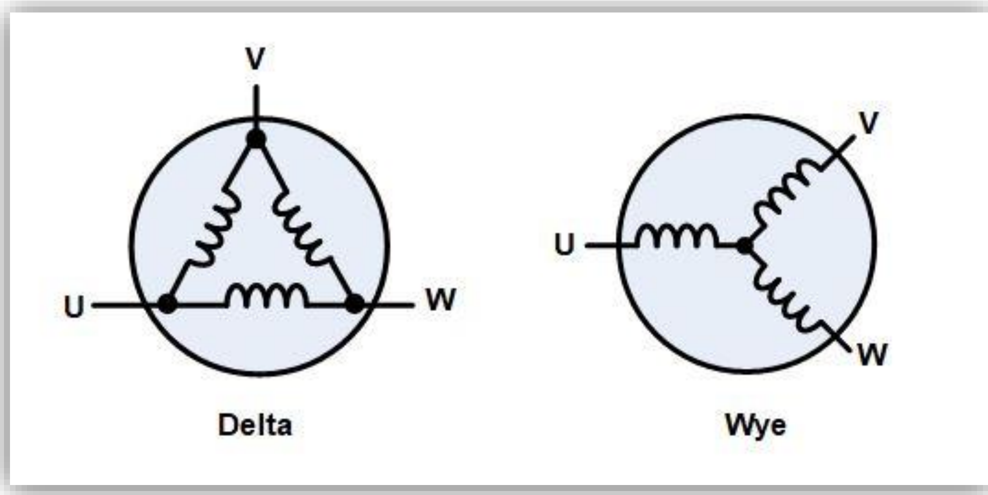


Figure 3.2.2.6.2: Brushless Motor

Three phases allow for the construction of motors with various magnetic configurations, or poles. The rotor only has one set of magnetic poles, North and South, making two-pole 3-phase motors the most basic type. In order to construct a motor with more poles, the rotor must have more magnetic sections, and the stator must have more windings. Although very high speeds are better achieved with fewer pole counts, higher pole counts can still result in improved performance.

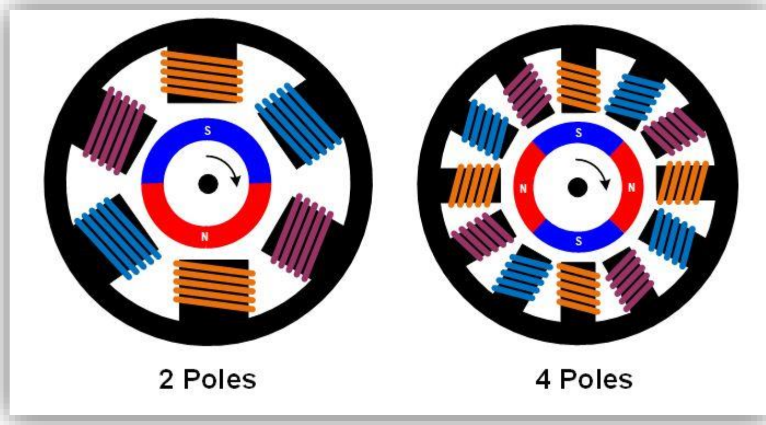


Figure 3.2.2.6.3: Brushless Motor

Each of the three phases must be able to be driven to either the input supply voltage or ground in order to run a three-phase brushless motor. Three "half bridge" driving circuits, each with two switches, are employed to achieve this. Bipolar transistors, IGBTs, or MOSFETs can be used as switches, depending on the needed voltage and current.

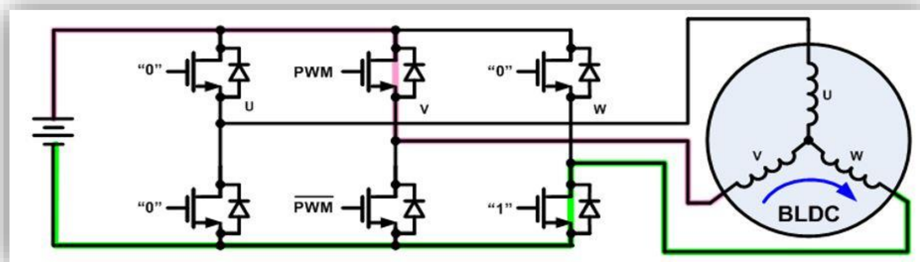


Figure 3.2.2.6.4: Brushless Motor

Three phase brushless motors are capable of being driven via a variety of methods. Trapezoidal, block, or 120-degree commutation are the three simplest. The commutation process utilized in a DC brush motor and trapezoidal commutation are relatively comparable. According to this plan, one of the three phases is always connected to ground, one is always left open, and the third phase is always driven to the supply voltage. When speed or torque control is required, the supply phase is often pulse width modulated. There is some variation in torque (referred to as torque ripple) when the motor rotates because the phases are quickly swapped at each commutation point while the rotor rotation is constant.

Other commutation techniques can be employed for better performance. Current is continuously driven through all three motor phases via sine, or 180-degree, commutation. Each phase of the drive electronics produces a sinusoidal current that is offset by 120 degrees from the other phase. This drive approach is frequently utilized for high performance or high efficiency drives because it reduces torque ripple, acoustic noise, and vibration.

The control electronics must understand the precise location of the magnets on the rotor in relation to the stator in order to rotate the field properly. Hall sensors affixed to the stator are frequently used to determine the location. The magnetic field of the rotor is detected by the Hall sensors as it rotates. The drive electronics use this information to send electricity to the stator windings in a specific order, which causes the rotor to spin.

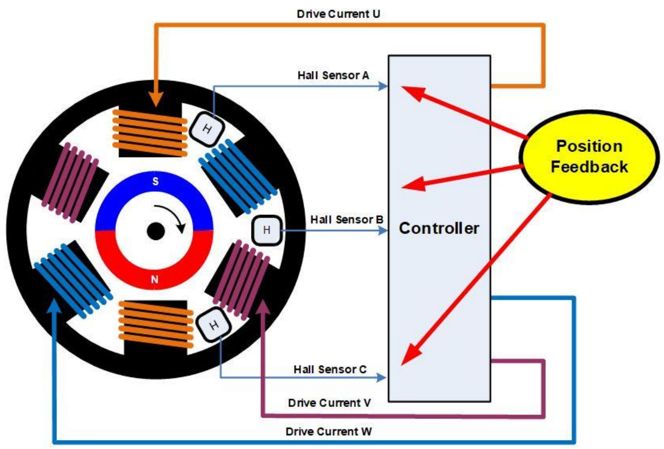


Figure 3.2.2.6.5: Brushless Motor

Trapezoidal commutation can be performed using three Hall sensors using only basic combinational logic, negating the need for complex control electronics. Other commutation techniques, such as sine commutation, call for slightly more complex control circuits and typically use a microprocessor.

There are several techniques that can be utilized to calculate the rotor position without sensors in addition to delivering position input utilizing Hall sensors. To determine the magnetic field in relation to the stator, the simplest method is to measure the back EMF during an undriven phase. Field Oriented Control, or FOC, is a more complex control algorithm that determines the location based on rotor currents and other factors. A fairly strong processor is often needed for FOC because many computations must be completed quickly. Of course, this is more expensive than a straightforward trapezoidal control approach.

7) ESC hw30a motor driver:

Figure 3.2.2.7.1 ESC hw30a motor driver

The HW30A Motor Speed Controller can be used with 4-10 NiMH/NiCd or 2-3 cell LiPo batteries. The BEC is functional with up to 3 LiPo cells. It can be used to control speed of Brushless DC motor (3 wires) with maximum up to 12Vdc.

Specification:

- Max Continues Current : 30A on 3 Cells
- Max Input Voltage: 12V
- BEC : 2A
- Input Voltage : 2-3 Lithium Polymer or 4-10 NiCd/NiMH
- Resistance: 0.0050 ohm
- FETs: 12 Lithium
- Cut OFF Voltage: 3.0V / cell
- Size: 45 x 24 x 9 mm
- Protection: 110 CPWM: 8KHzMax Rotation Speed 20,000 RPM for 14 pole motor

8) Raspberry pi HQ Camera:

Figure 3.2.2.8.1: Raspberry pi HQ Camera

The Raspberry Pi High-Quality Camera is the latest camera accessory from Raspberry Pi. It offers higher resolution and sensitivity (approximately 50% greater area per pixel for improved low-light performance) than the existing Camera Module v2.
The package comprises a circuit board carrying a Sony IMX477 sensor, an FFC (Flexible Flat Cable) cable for connection to a Raspberry Pi computer, a milled aluminum lens mount with integrated tripod mount, and focus adjustment ring, and a C- to CS-mount adapter. This is the first official HQ (High Quality) camera module from Raspberry Pi.

Features:
•Image Sensor:
◦SONY IMX477R stacked, black-illuminated sensor
◦12.3 Megapixels
◦7.9mm sensor diagonal
◦1.55um x 1.55 um pixel size

•Output: RAW12/10/8, COMP8
•Back Focus: Adjustable (12.5mm – 22.4mm)
•Lens standards: C-mount or CS-mount (C-CS-mount adapter included and mounted on the camera by default)
•No focusing lens included, but it is a MUST to have a lens, so get it separately:
◦Raspberry Pi 6mm lens
◦Raspberry Pi 16mm lens

•IR filter: integrated
•Ribbon cable/FFC length: 200mm
•Tripod mount: 1/4″ – 20
•Compliance:
◦FCC 47 CFR Part 15, Subpart B, Class B Digital Device
◦Electromagnetic Compatibility Directive (EMC) 2014/30/EU
◦Restriction of Hazardous Substances (RoHS) Directive 2011/65/EU
•Connects to the Raspberry Pi board via the FFC/Ribbon cable (included)
•Size: 38mm x 38mm x 18.43mm

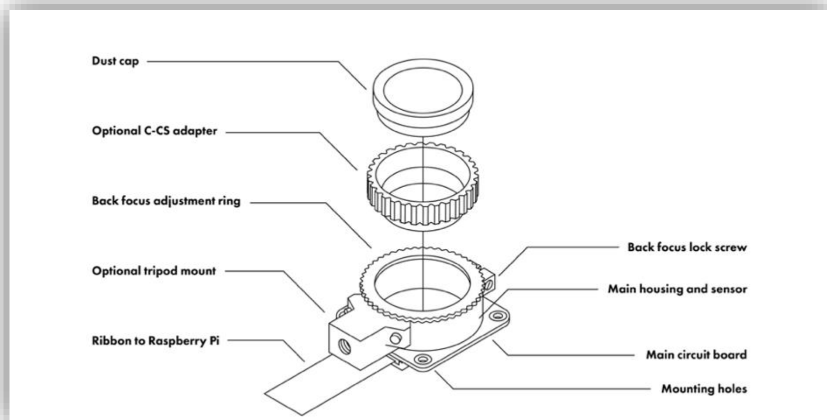


Figure 3.2.2.8.2: Raspberry pi HQ Camera

9) GPS M8N module:

Figure 3.2.2.9: GPS M8N module

The M8N GPS is the primary CAN protocol used by the ArduPilot and PX4 projects for communication with CAN peripherals. It is an open protocol with open communication, specification, and multiple open implementations. It has the STM32G4 Processor and adopted the Drone CAN protocol for communication, making it more reliable and better in dealing with electromagnetic interference compare to serial connection. It does not occupy any serial port of the flight controller, and different CAN devices can be connected to the same CAN bus via a CAN splitter board.

10)Battery:

Figure 3.2.2.10: Battery

Battery: LiPo batteries should be used to power drones. Because LiPo batteries produce power quicker, store more power, and have a longer life than their NiCad elder siblings, they are much superior than NiCad batteries

11) Quadcopter frame:

The framework that connects all of the components. One of the most important parts of a quadcopter is the frame, which supports the motors and other electronics and shields them from vibrations. Extreme precision must be used in its creation. They need to be built to be both strong and lightweight.

Figure 3.2.2.11.1: Quadcopter frame

Propellers: Are mechanical devices that convert rotational motion into linear propulsion. Drone propellers produce lift for the aircraft by spinning and producing airflow, resulting in a pressure differential between the propeller's top and bottom surfaces. This causes a mass of air to accelerate in one direction, creating lift that counteracts gravity.

The drone can hover, rise, descend, or change its yaw, pitch, and roll by changing the speed of these propellers.

Figure 3.2.2.11.2: Propellers

12) RC transmitter and receiver:

Figure 3.2.2.12: RC transmitter and receiver

As comparison to Wi-Fi or Bluetooth, RC remote (transmitters) enables you to fly your drone over a considerably wider area.

Drones often come with a certain RC transmitter, however since RC transmitters can be switched out, you can also use another RC transmitter if you'd prefer. Be that your drone can communicate with the third-party RC transmitter you plan to use, or that you can install the necessary receiver in your drone.

13) PM02 V3-12S Power Module:

Figure 3.2.2.13: PM02 V3-12S Power Module

With the Power Module, SmartAP can be supplied with clean power through a LiPo battery, as well as monitor current consumption and battery voltage over a 6- positon cable. Using up to a 6S LiPo battery, it provides 5.3V / 3A of DC power.

XT60 connectors are built into the Power Module and covered in shrink tubing for protection.

This device measures the battery voltage as well as the battery current. By calculating the remaining battery power and estimating the remaining flight time, SmartAP detects the battery voltage and the battery current. A stable power supply is also provided by the module's BEC feature.

Specifications:

• Below are typical limits, but it is best to confirm directly with the vendor:

• Maximum input voltage of 18V (4S lipo)

• Maximum of 90 Amps (but only capable of measuring up to 60 Amps)

• Provides 5.37V and 2.25Amp power supply to the autopilot

3.2.3)- Software Specifications:

Tools	Description	Image
OpenCV	It is a big open-source library dedicated to machine learning, image processing, and computer vision. It assists a broad mixture of programming languages like C++, Python, and Java, etc. It's able of analyzing videos and pictures to detect faces, objects, along with human handwriting. [34]	*Figure 3.2.3.1 OpenCV software*
Anaconda Navigator	It is a free or paid special distro program whose main purpose is to facilitate the management of certain libraries. The libraries we are talking about in Ria are libraries that specialize in dealing with Big Data, Scientific Computing Applications, and other scientific libraries. Anaconda is famous for data science, data processing, and machine learning. If the question is why we don't use the libraries we need by going to their source and installing them directly using a tool pip .or one of the other methods.	*Figure 3.2.3.2 Anaconda Navigator*
Spyder	It is an integrated scientific development environment designed for scientific purposes, engineers, and also students like us, where it provides a unique combination of advanced editing, analysis, and debugging, and one of its most important advantages is that it allows the implementation of the code in a fragmented way to enable us to detect errors with ease. And contribute to the process of creating programs.	*Figure 3.2.3.3 Spyder*

Python	Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python is dynamically-typed and garbage-collected. It supports multiple programming paradigms, including structured, object-oriented and functional programming. Python is a computer programming language often used to build websites and software, automate tasks, and conduct data analysis. Python is a general-purpose language, meaning it can be used to create a variety of different programs and isn't specialized for any specific problems.	***Figure 3.2.3.4 Python***

3.3) System Block Diagrams:

3.3.1)- The Block Diagram of the system

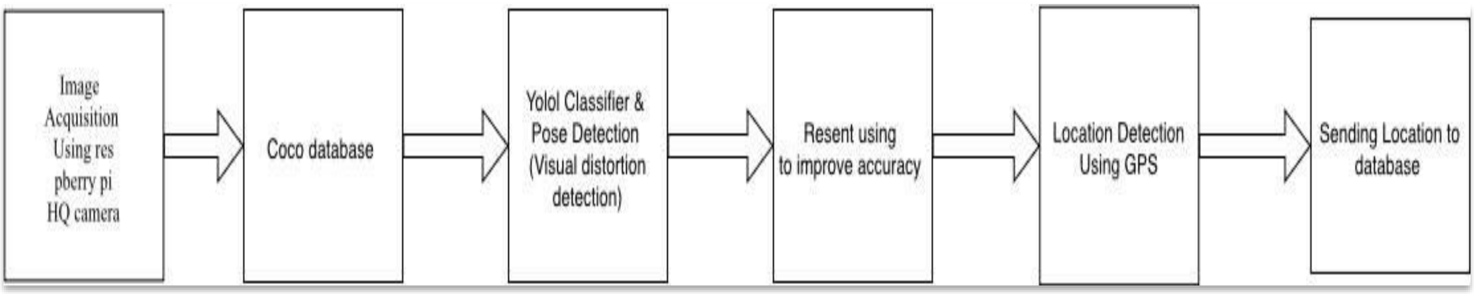


Figure 3.3.1: The Block Diagram of the system

3.3.2)- Dataflow Diagram of the system

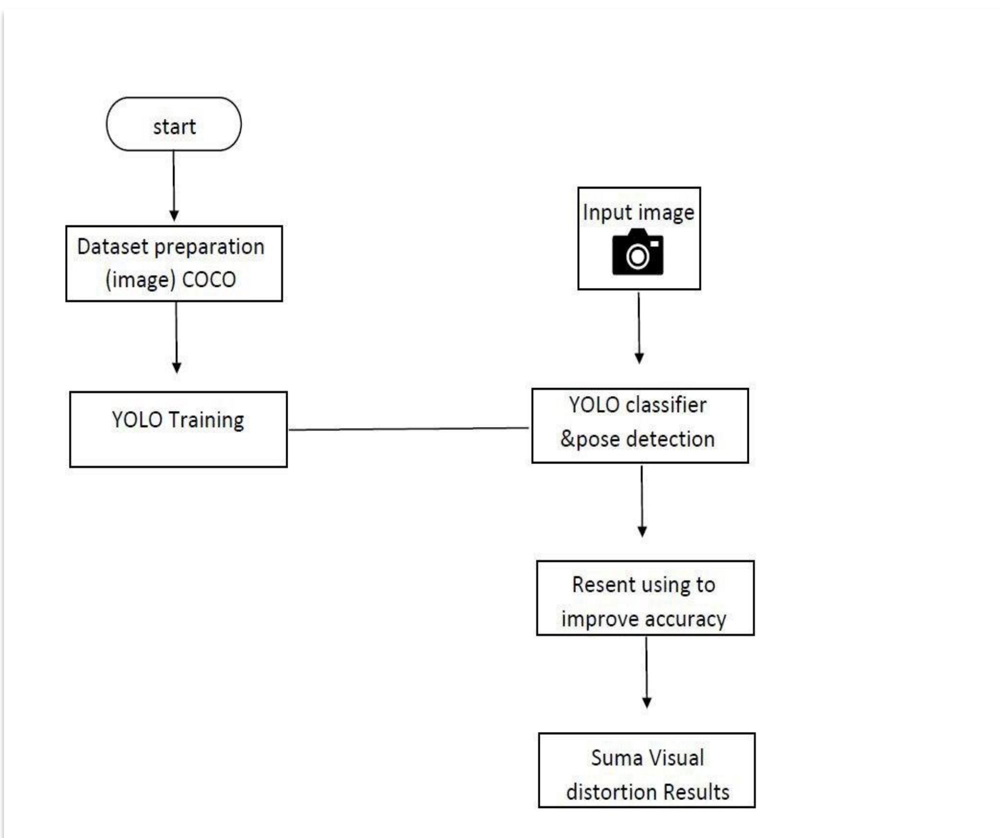


Figure 3.3.2: Dataflow Diagram of the system

3.3.3) - Circuit Diagram of Digital video transmission system using openHd:

A)-Pi video transmitter:

- Raspberry pi 3 model B
- Antenna
- Raspberry pi HQ Camera
- Power module: 14v DC to 5v DC converter to power the system

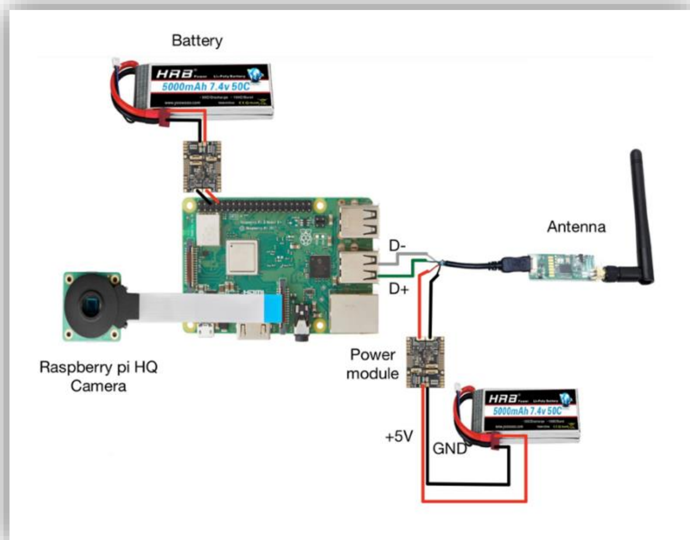


Figure 3.3.1.1: Pi video transmitter

B)- Pi video receiver:

- Raspberry pi 4 model B
- long rang wifi adapter.

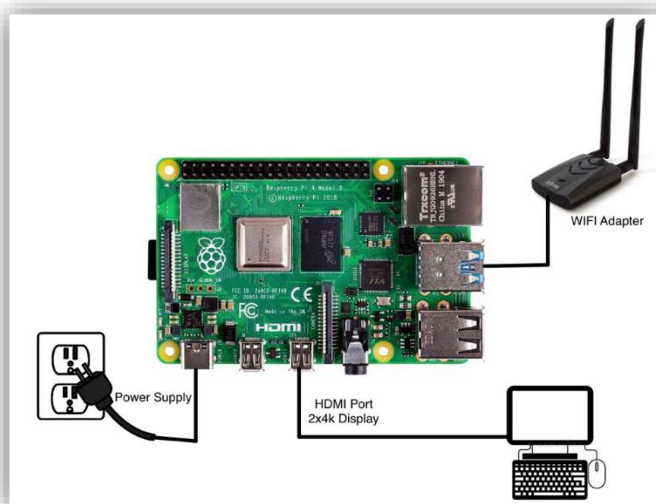


Figure 3.3.1.2: Pi video receiver

3.3.4)- Circuit Diagram of Flight Controller:

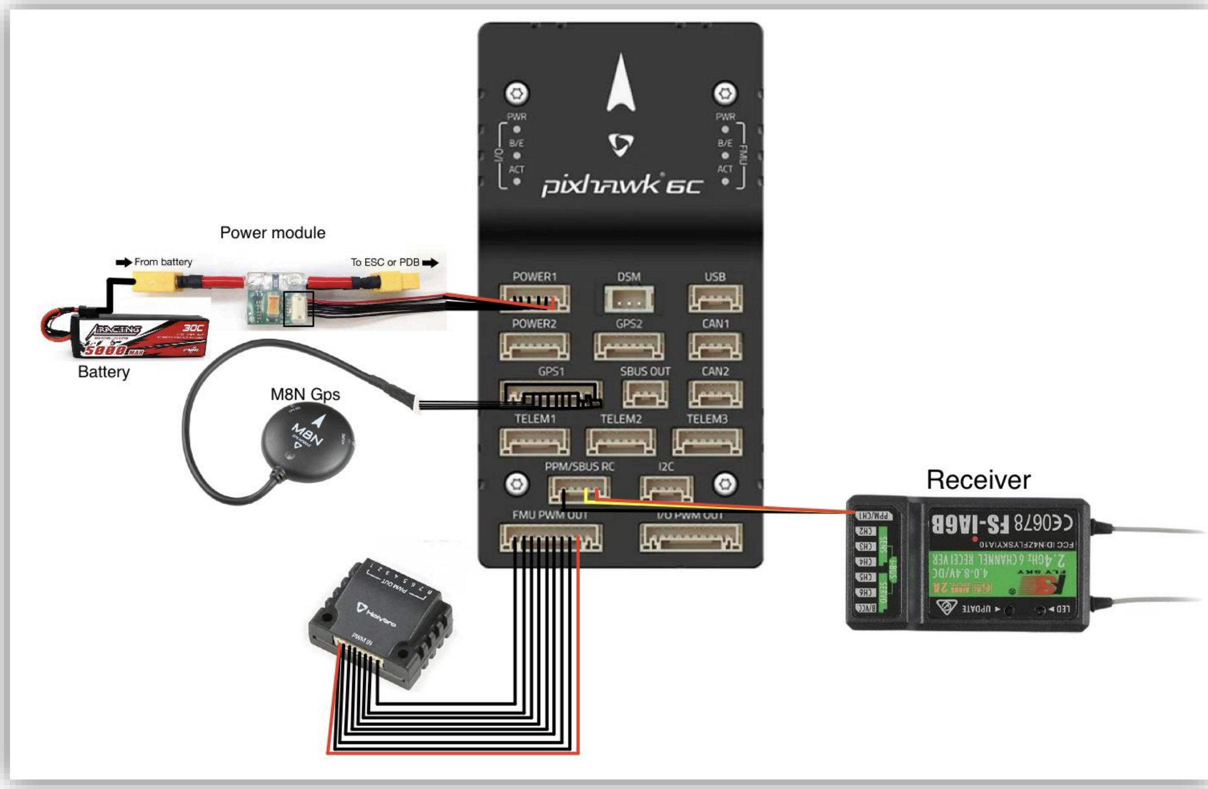


Figure 3.3.4: Circuit Diagram of Flight Controller

3.3.5)- Circuit Diagram of Connect ESC (Electronic Speed Controller) and Motor To PDB (Power distribution board)

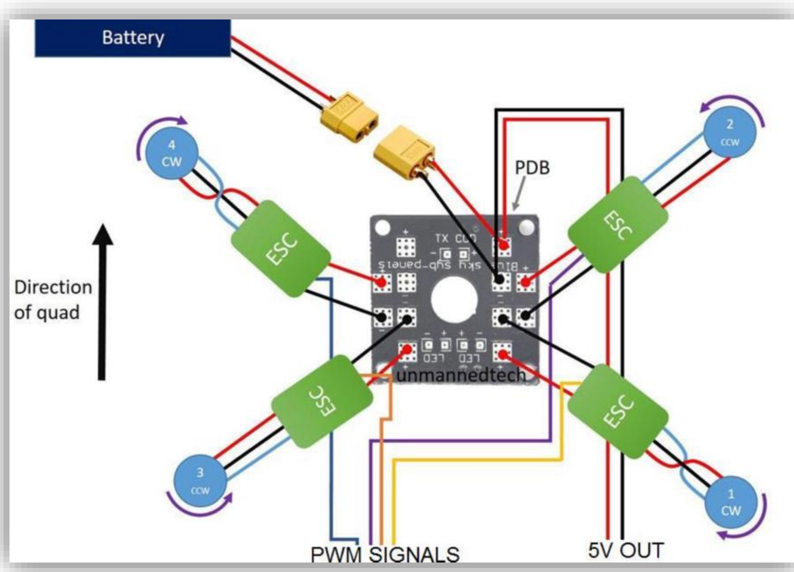


Figure 3.3.5: Circuit Diagram of connect ESC and Motor

Chapter 4: Simulations and Results

4.1) Introduction

In this chapter we have done the simulation via python is among the most widely used programming languages that developers use in the present. Guido Van Rossum created it in 1991, and since its beginning, it has been among the most popular languages alongside C++, Java, and others.

Python has taken a large lead in our quest to determine the best programming language for AI or neural networks.

4.2) General flowchart:

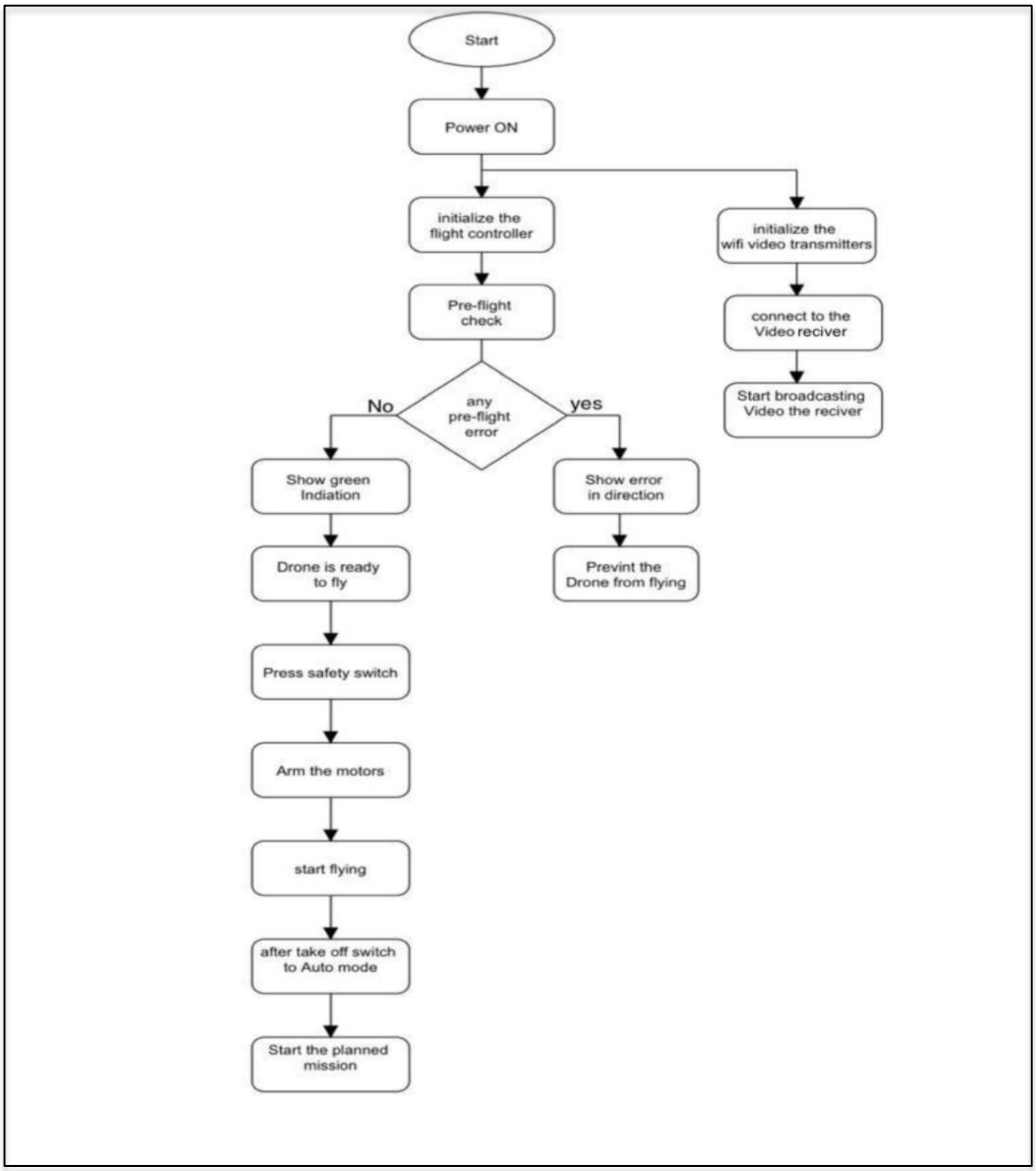


Figure 4.2.1: Display transmission drone Flowchart

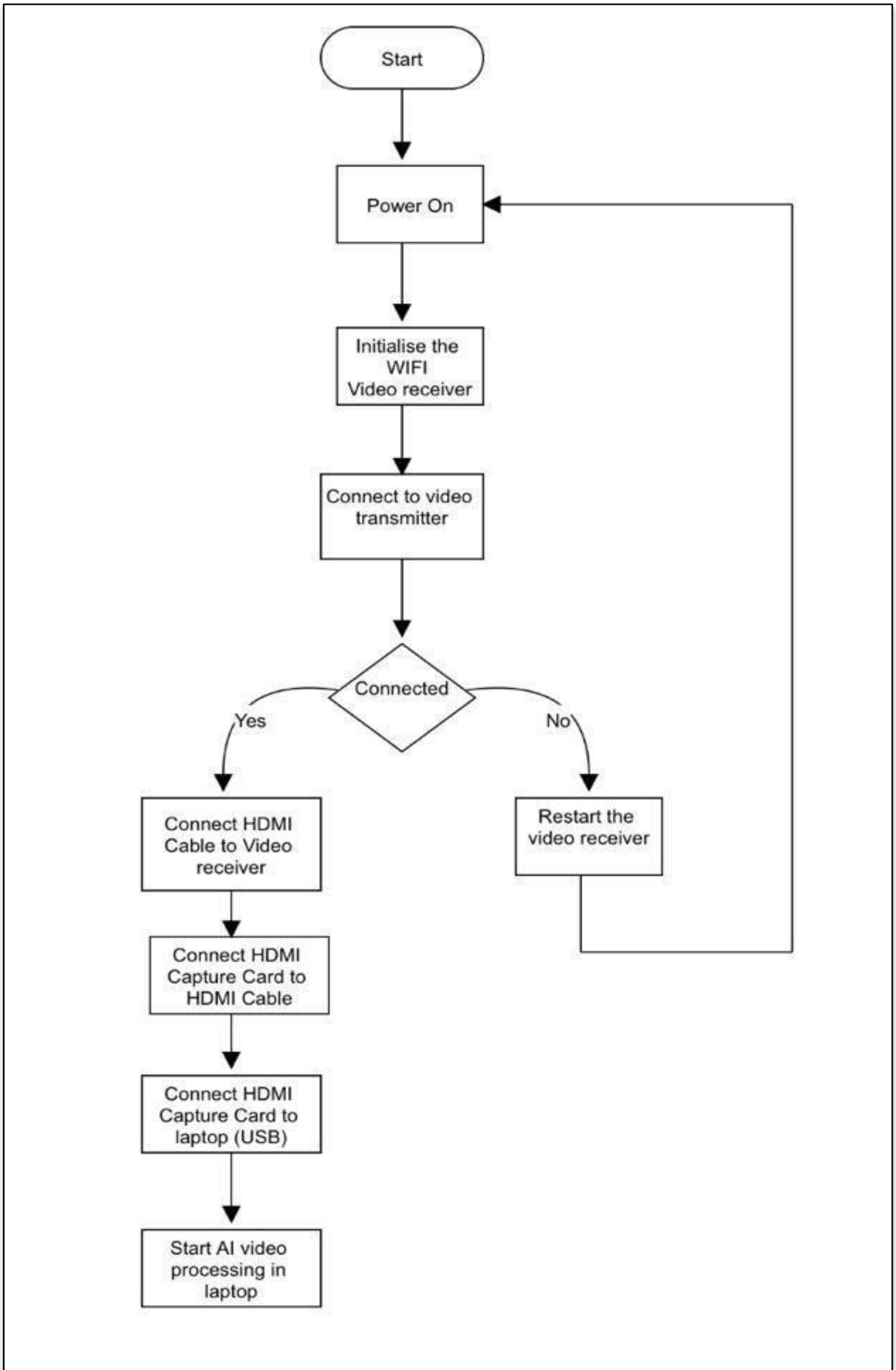


Figure 4.2.2: Display and process the video Flowchart

4.3) Circuit Diagram

Circuit Diagram:

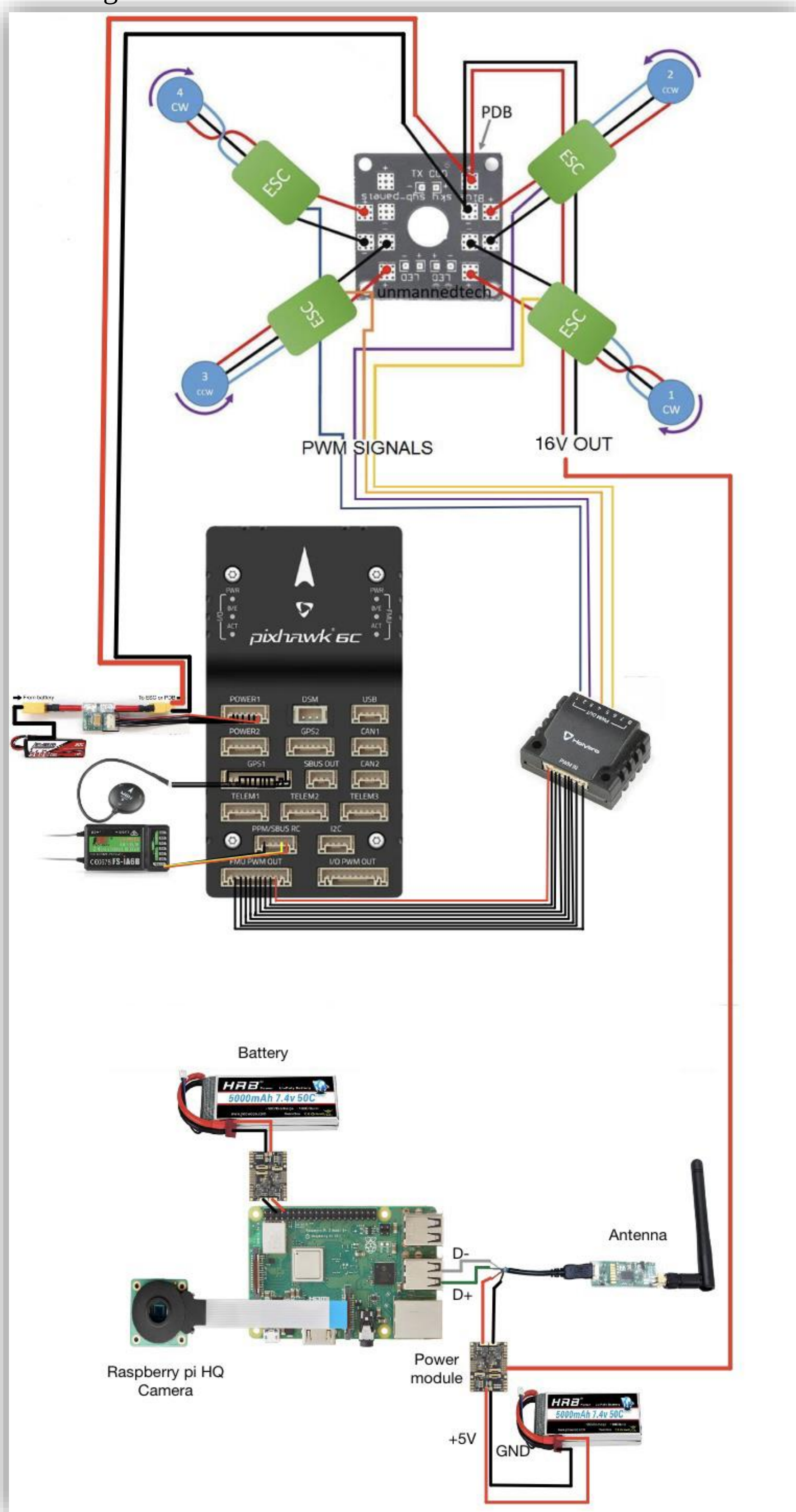


Figure 4.3.1: Circuit Diagram

4.4) Algorithms used in the code:

4.4.1 Common Objects in Context (COCO)

Since the picture dataset was produced with the intention of advancing image recognition, it is known as COCO, or Common Objects in Context. State-of-the-art neural networks are mostly used in the demanding, high-quality visual datasets included in the COCO dataset for computer vision. For instance, COCO is frequently used to evaluate algorithms and assess how well they perform in real-time object detection. Modern libraries for neural networks automatically interpret the COCO dataset's format. Can develop object detection models using the COCO dataset. 80 different types of items are represented by bounding box coordinates in the dataset, which may be used to train models that can identify bounding boxes and categorize objects in photos.

Figure 4.4.1.1 Common Objects in Context

Using pre-defined classes, these items are tagged. An extremely common technique in computer vision is labeling, which is sometimes known as image annotation. In contrast to existing object recognition datasets that have concentrated on

1) Picture classification

2) Object bounding-box localization

3) Semantic pixel-level segmentation

4) Segmenting individual object instances.

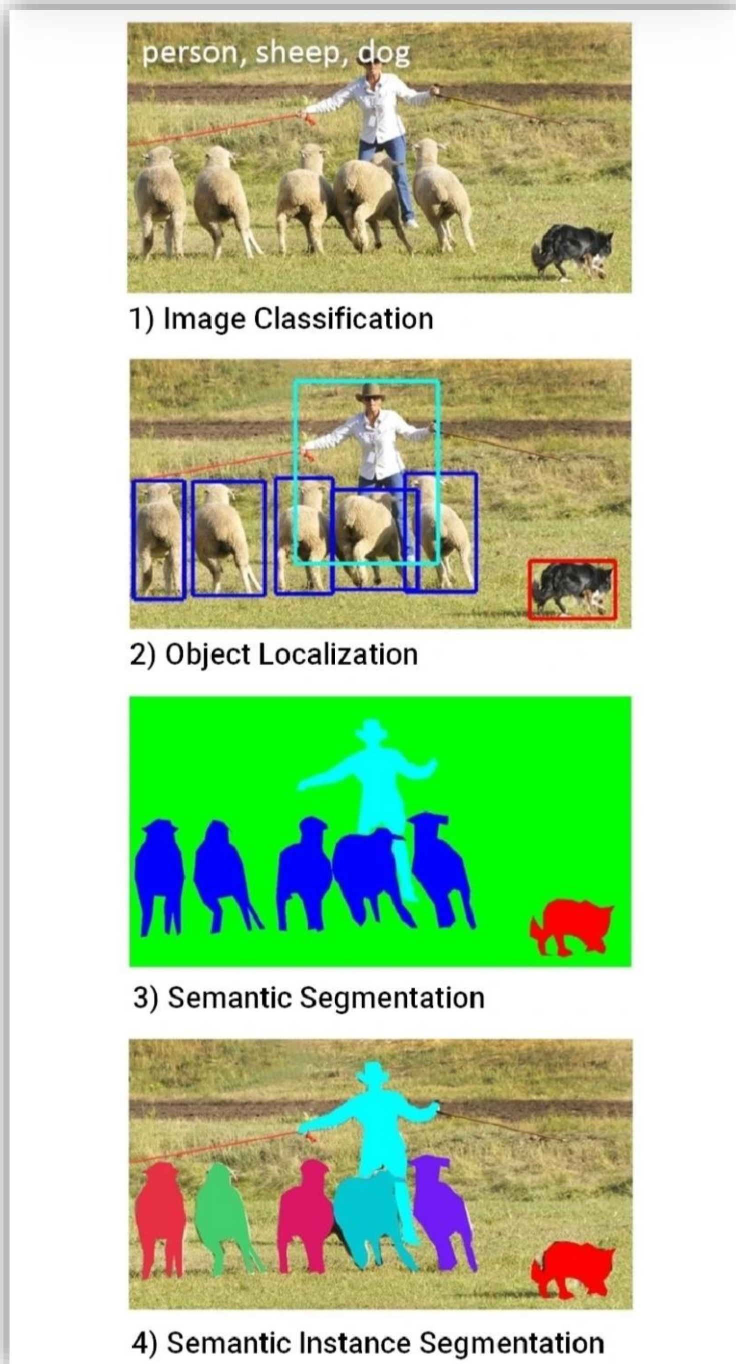


Figure 4.4.1.2 Common Objects in Context

We applied this algorithm in our project (drone discover) so that it recognizes the offending images and classifies the image, for example, garbage (garbage bags, bottles, etc.) and then puts a box surrounding the object, segmenting the object by pixels, segmenting the individual object.

4.4.2) You Only Look Once (YOLO):

YOLO is an algorithm that uses neural networks to provide real-time object detection. This algorithm is popular because of its speed and accuracy.

This algorithm identifies and finds different things in a picture (in real-time). Convolutional neural networks (CNN) are used by the YOLO algorithm to recognize items instantly. The approach just needs one forward propagation through a neural network to detect objects.

We used YOLO algorithm because of the following reasons:

- Speed: This algorithm improves the speed of detection because it can predict objects in real-time.
- High accuracy: YOLO is a predictive technique that provides accurate results with minimal background errors.
- Learning capabilities: The algorithm has excellent learning capabilities that enable it to learn the representations of objects and apply them in object detection.

YOLO algorithm works using the following three techniques:

- Residual blocks
- Bounding box regression
- Intersection Over Union (IOU)

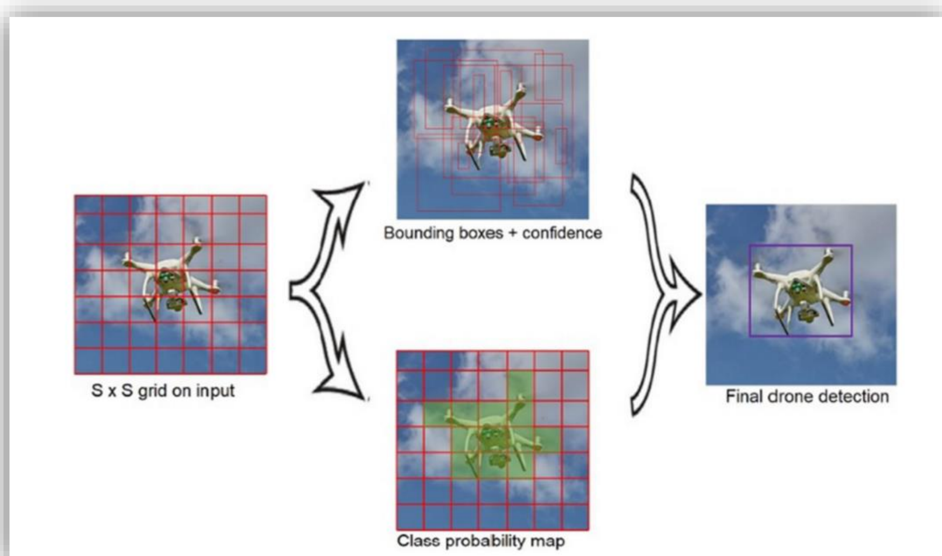


Figure 4.4.2.1: You Only Look Once

4.4.3) Residual Network (ResNet):

We used **ResNet** because it provides an innovative solution to the vanishing gradient problem.

What is ResNet?

is a deep learning model used for computer vision applications. It is a Convolutional Neural Network (CNN) architecture.

How it works?

ResNet stacks multiple identity mappings, skips those layers, and reuses the activations of the previous layer. Skipping speeds up initial training by compressing the network into fewer layers .

Then, when the network is retrained, all layers are expanded and the remaining parts of the network—known as the residual parts—are allowed to explore more of the feature space of the input image

Neural networks are trained through a backpropagation process that relies on gradient descent, shifting down the loss function and finding the weights that minimize it. If there are too many layers, repeated multiplications will eventually reduce the gradient until it “disappears”, and performance saturates or deteriorates with each layer added

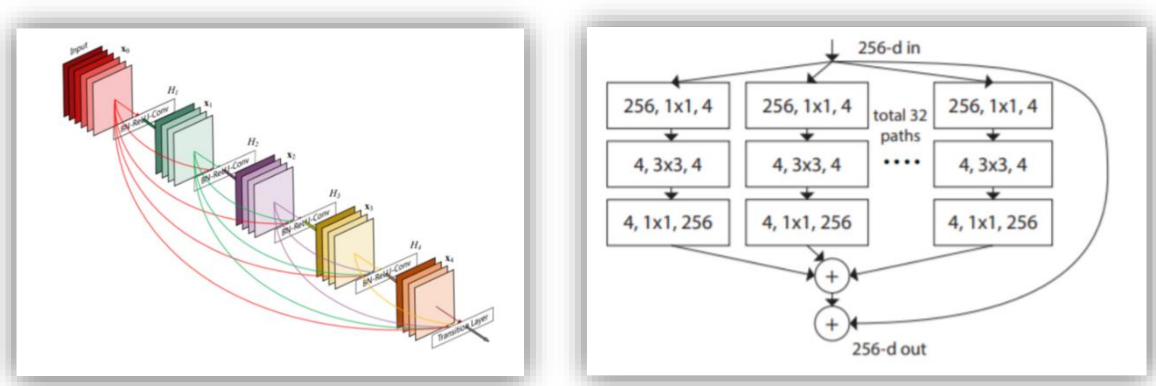


Figure 4.4.3.1 Residual Network (ResNet)

4.4.4) Training the model:

4.4.4.1 Training YOLOv5 Object, Detector on a Custom Dataset

With the help of Deep Learning, we all know that the field of Computer Vision has proliferated in the last decade. As a result, so many prevalent computer vision problems like image classification, object detection, and segmentation having real industrial use-case started to achieve accuracy like never before.

we did a custom object detection on a dataset consists of 131 images

```
Transferred 349/349 items from yolov5s.pt
ptimizer: SGD(lr=0.01) with parameter groups 57 weight(decay=0.0), 60 weight(decay=0.0005), 60 bias
rain: Scanning D:\python\datasets\coco128\labels\train2017... 131 images, 0 backgrounds, 0 corrupt: 100%|          | 1
rain: WARNING  Cache directory D:\python\datasets\coco128\labels is not writeable: [WinError 183] Cannot create a file
when that file already exists: 'D:\\python\\datasets\\coco128\\labels\\train2017.cache.npy' -> 'D:\\python\\datasets\\co
co128\\labels\\train2017.cache'
val: Scanning D:\python\datasets\coco128\labels\train2017... 131 images, 0 backgrounds, 0 corrupt: 100%|          | 131
val: WARNING  Cache directory D:\python\datasets\coco128\labels is not writeable: [WinError 183] Cannot create a file wh
en that file already exists: 'D:\\python\\datasets\\coco128\\labels\\train2017.cache.npy' -> 'D:\\python\\datasets\\coco
128\\labels\\train2017.cache'
```

Figure 4.4.4.1 Training YOLOv5

4.4.5.1) Training setting (Annotations):

```
path: ../datasets/coco128  # dataset root dir
train: images/train2017  # train images (relative to 'path') 128 images
val: images/train2017  # val images (relative to 'path') 128 images
test:  # test images (optional)

# Classes
names:
  0: garbage
  1: cans
  2: trash
  3: glass
  4: metals
  5: grafitti
  6: illegal
  7: water
```

Figure 4.4.5.1 Training YOLOv5

Thing that are specified in the pictures:

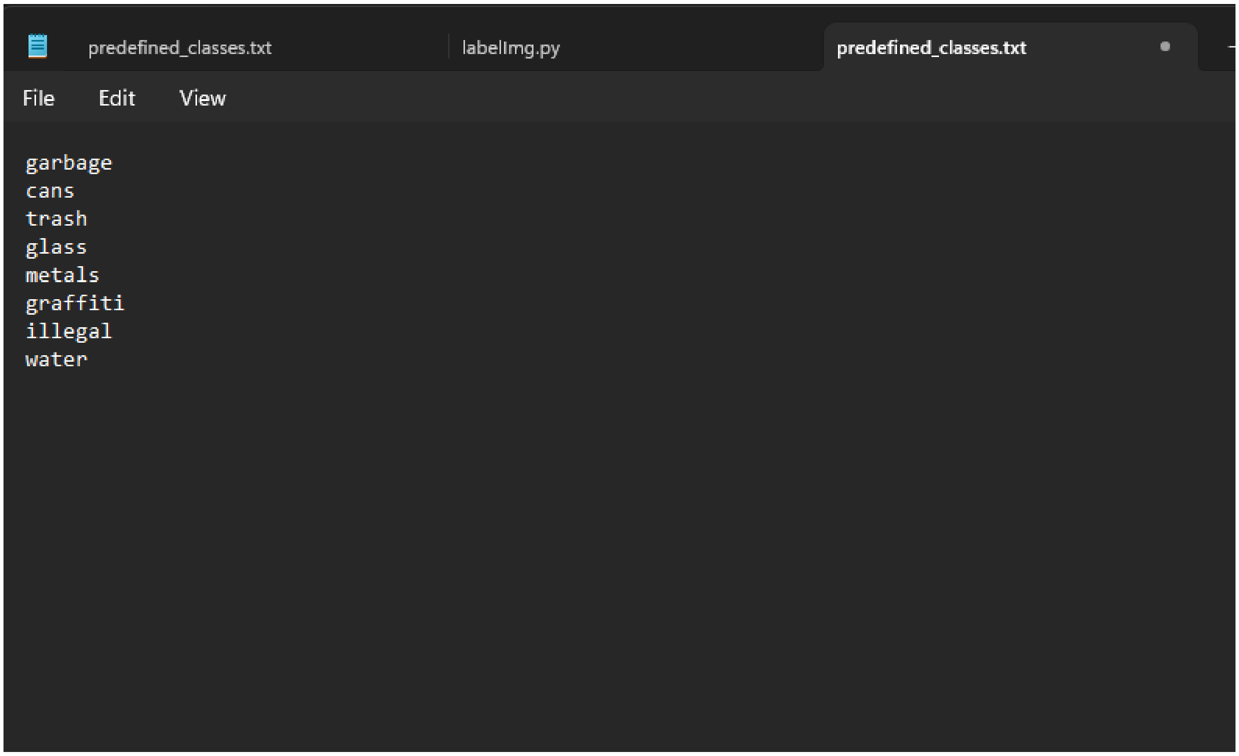


Figure 4.4.5.2: Training YOLOv5

4.4.6.1) labellmg:

Anaconda prompet

Figure 4.4.6.1: Anaconda prompet

Trash (Rectangles are selected on the images)

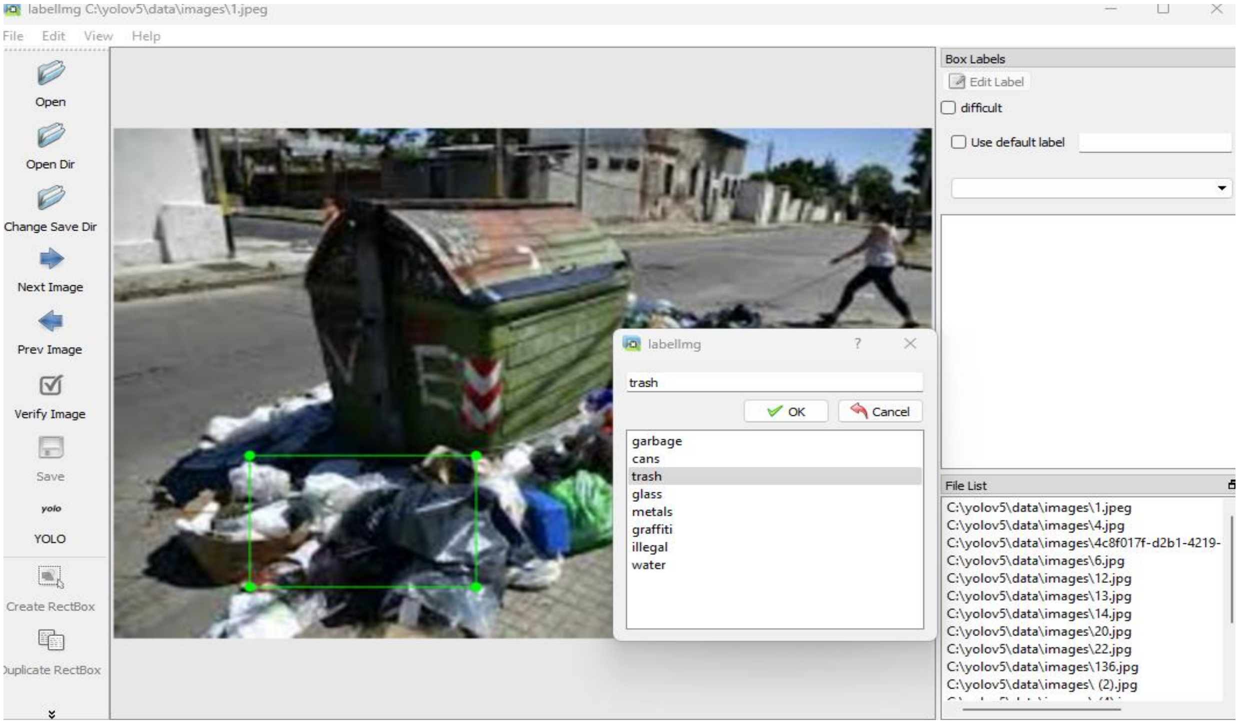


Figure 4.4.6.2: Trash

illegal

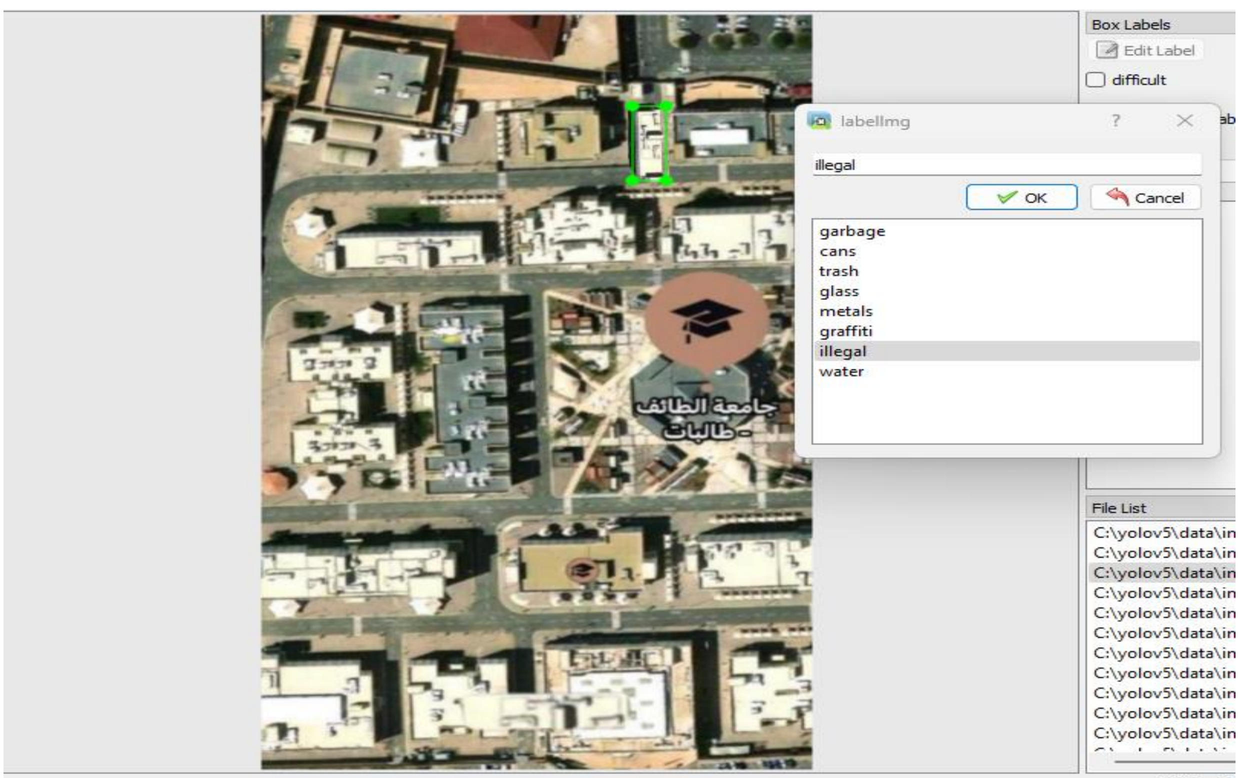


Figure 4.4.6.3: illegal

4.4.7.1) Train:

```python
import math
import os
import random
import subprocess
import sys
import time
from copy import deepcopy
from datetime import datetime
from pathlib import Path

import numpy as np
import torch
import torch.distributed as dist
import torch.nn as nn
import yaml
from torch.optim import lr_scheduler
from tqdm import tqdm

FILE = Path(__file__).resolve()
ROOT = FILE.parents[0]  # YOLOv5 root directory
if str(ROOT) not in sys.path:
    sys.path.append(str(ROOT))  # add ROOT to PATH
ROOT = Path(os.path.relpath(ROOT, Path.cwd()))  # relative

import val as validate  # for end-of-epoch mAP
from models.experimental import attempt_load
from models.yolo import Model
from utils.autoanchor import check_anchors
from utils.autobatch import check_train_batch_size
from utils.callbacks import Callbacks
from utils.dataloaders import create_dataloader
from utils.downloads import attempt_download, is_url
from utils.general import (LOGGER, TQDM_BAR_FORMAT, check_amp, check_dataset, check_file, check_git_inf
                           check_git_status, check_img_size, check_requirements, check_suffix, check_ya
                           get_latest_run, increment_path, init_seeds, intersect_dicts, labels_to_class
```

Figure 4.4.7.1: Train

train.py

```
Console 1/A

                 from  n    params  module
arguments
  0                -1  1      3520  models.common.Conv
[3, 32, 6, 2, 2]
  1                -1  1     18560  models.common.Conv
[32, 64, 3, 2]
  2                -1  1     18816  models.common.C3
[64, 64, 1]
  3                -1  1     73984  models.common.Conv
[64, 128, 3, 2]
  4                -1  2    115712  models.common.C3
[128, 128, 2]
  5                -1  1    295424  models.common.Conv
[128, 256, 3, 2]
  6                -1  3    625152  models.common.C3
[256, 256, 3]
  7                -1  1   1180672  models.common.Conv
[256, 512, 3, 2]
  8                -1  1   1182720  models.common.C3
[512, 512, 1]
  9                -1  1    656896  models.common.SPPF
[512, 512, 5]
```

Figure 4.4.7.2: Train

4.4.7.2) Train_batch (visual distortion):

Figure 4.4.7.3 shows some sample images from the dataset with ground-truth bounding boxes annotated in green and red color etc.

Figure 4.4.7.3 Train batch

exp (1).zip

4.4.8.1) Training the model:

4.4.8.2 Training results

Next, we look at the results.png, which comprises training and validation loss for bounding box, objectness, and classification. It also has the metrics: precision, recall.

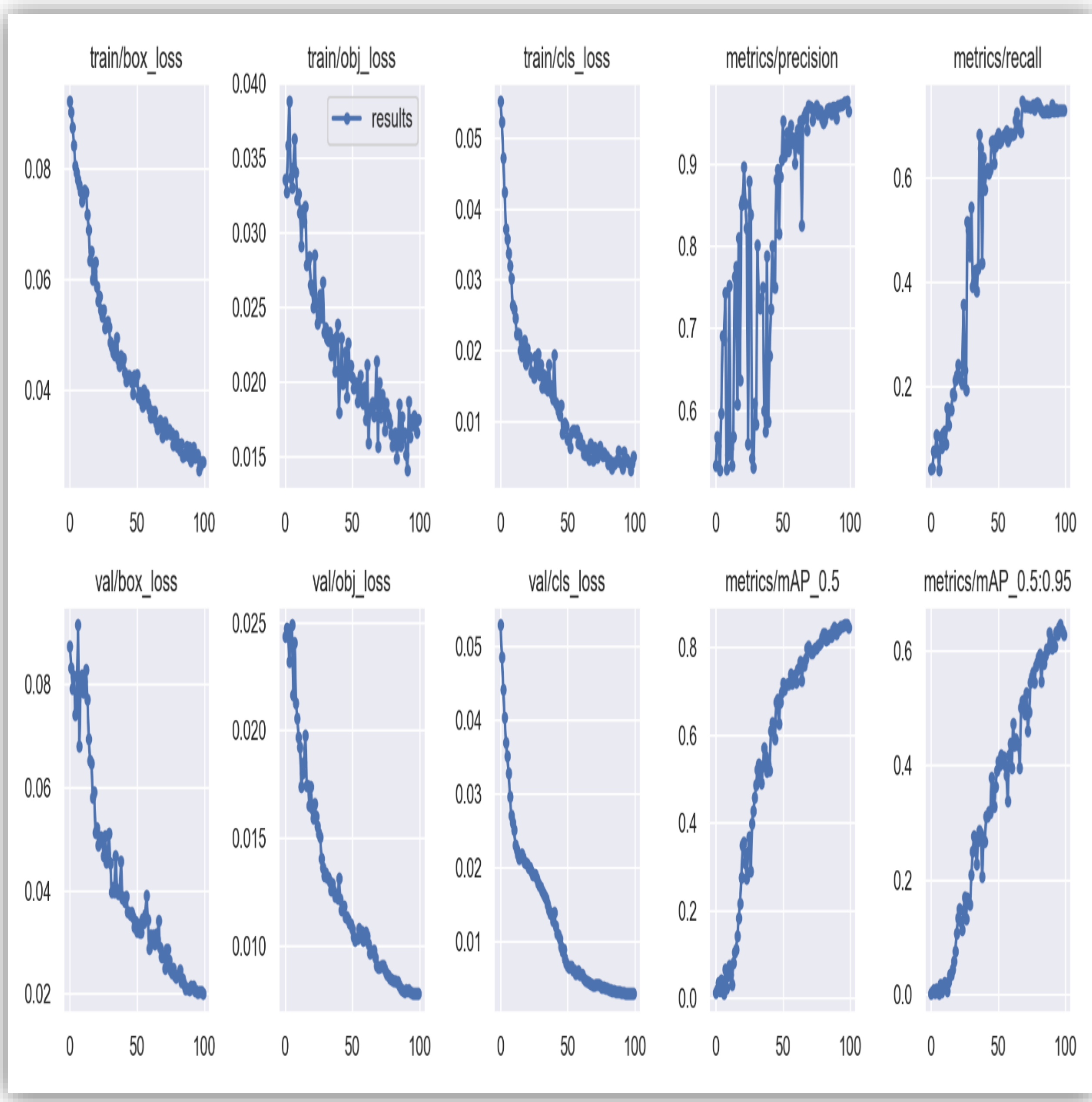


Figure 4.4.8.2: Results.png

Labels:

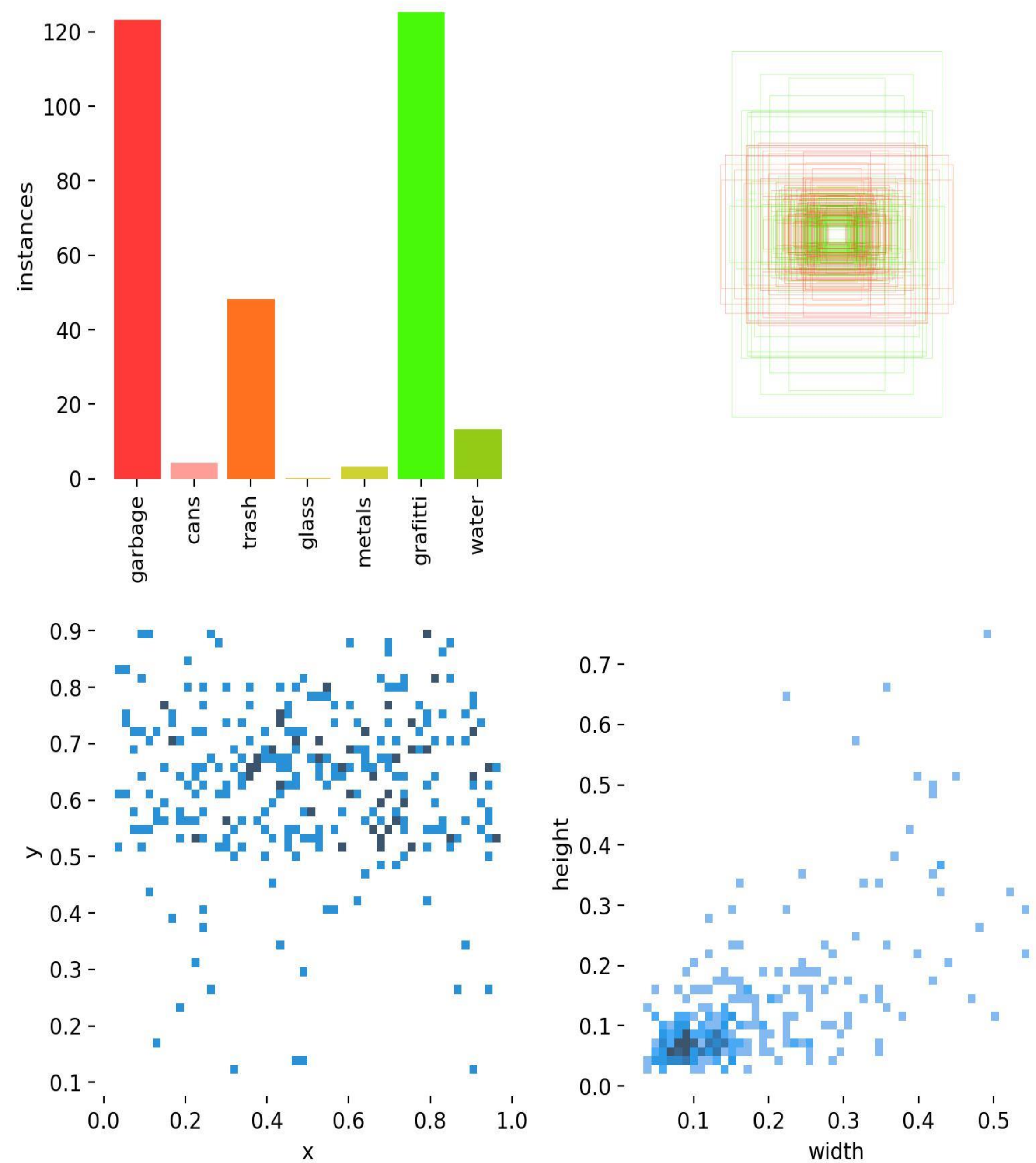


Figure 4.4.8.3: Labels

labels_correlogram:

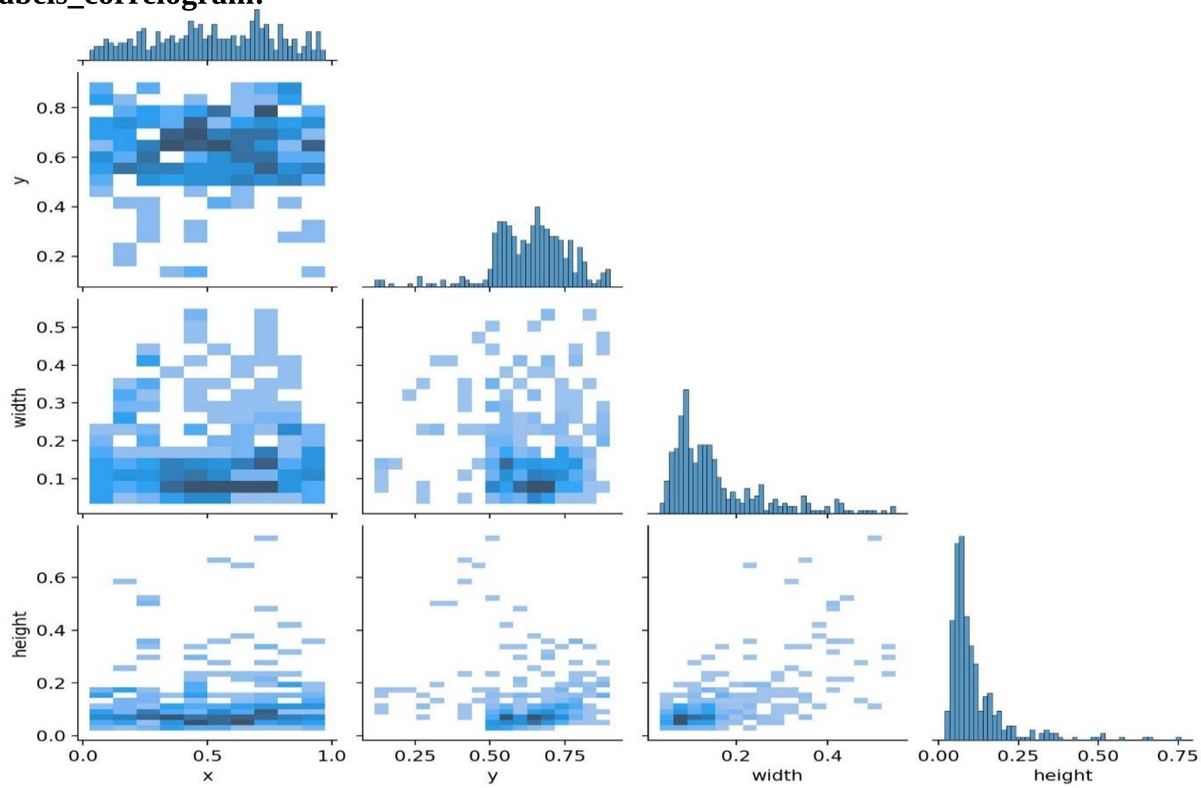


Figure 4.4.8.4: labels_correlogram

Confusion_matrix:

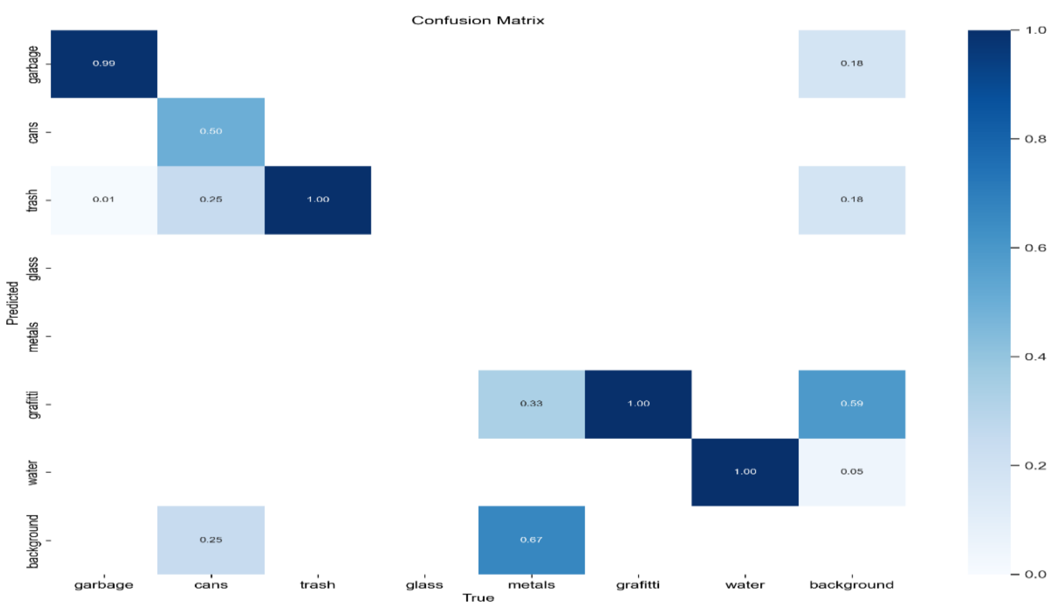


Figure 4.4.8.5: Confusion_matrix

Curve:

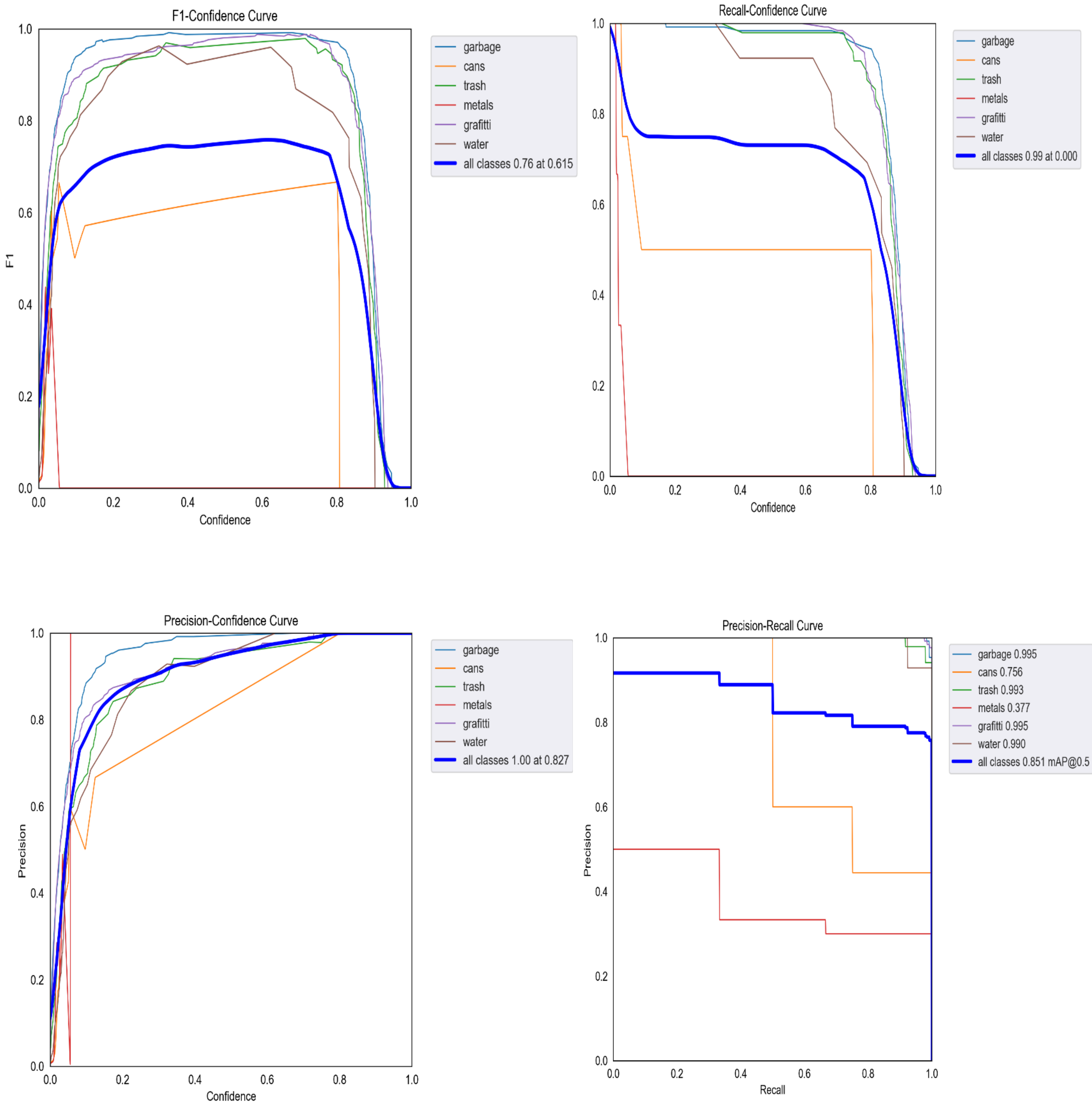


Figure 4.4.8.5: curve

4.4.9.1) Train_batch (<u>illegal buildings</u>):

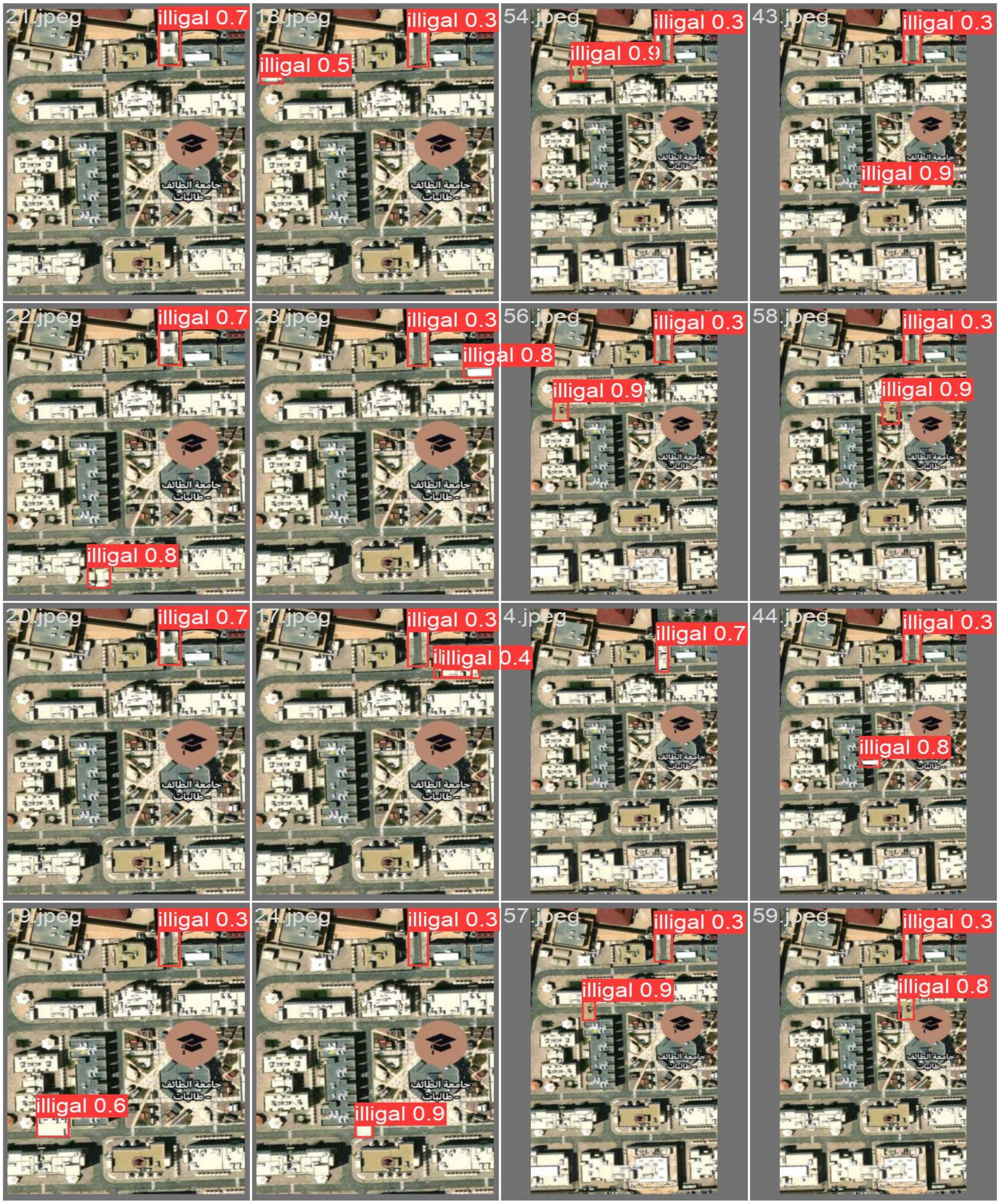


Figure 4.4.9.2: Confusion

results (2).zip

4.4.10.1) The code detection:

```python

import argparse
import os
import platform
import sys
from pathlib import Path

import torch

FILE = Path(__file__).resolve()
ROOT = FILE.parents[0]  # YOLOv5 root directory
if str(ROOT) not in sys.path:
    sys.path.append(str(ROOT))  # add ROOT to PATH
ROOT = Path(os.path.relpath(ROOT, Path.cwd()))  # relative

from models.common import DetectMultiBackend
from utils.dataloaders import IMG_FORMATS, VID_FORMATS, LoadImages, LoadScreenshots, LoadStreams
from utils.general import (LOGGER, Profile, check_file, check_img_size, check_imshow, check_requirements, colorstr, cv2,
                           increment_path, non_max_suppression, print_args, scale_boxes, strip_optimizer, xyxy2xywh)
from utils.plots import Annotator, colors, save_one_box
from utils.torch_utils import select_device, smart_inference_mode


@smart_inference_mode()
def run(
        weights=ROOT / 'yolov5s.pt',  # model path or triton URL
        source=ROOT / 'data/images',  # file/dir/URL/glob/screen/0(webcam)
        data=ROOT / 'data/coco128.yaml',  # dataset.yaml path
        imgsz=(640, 640),  # inference size (height, width)
        conf_thres=0.25,  # confidence threshold
        iou_thres=0.45,  # NMS IOU threshold
        max_det=1000,  # maximum detections per image
        device='',  # cuda device, i.e. 0 or 0,1,2,3 or cpu
        view_img=False,  # show results
        save_txt=False,  # save results to *.txt
        save_conf=False,  # save confidences in --save-txt labels
        save_crop=False,  # save cropped prediction boxes
        nosave=False,  # do not save images/videos
        classes=None,  # filter by class: --class 0, or --class 0 2 3
        agnostic_nms=False,  # class-agnostic NMS
        augment=False,  # augmented inference
```

Figure 4.4.10.1 The code detection

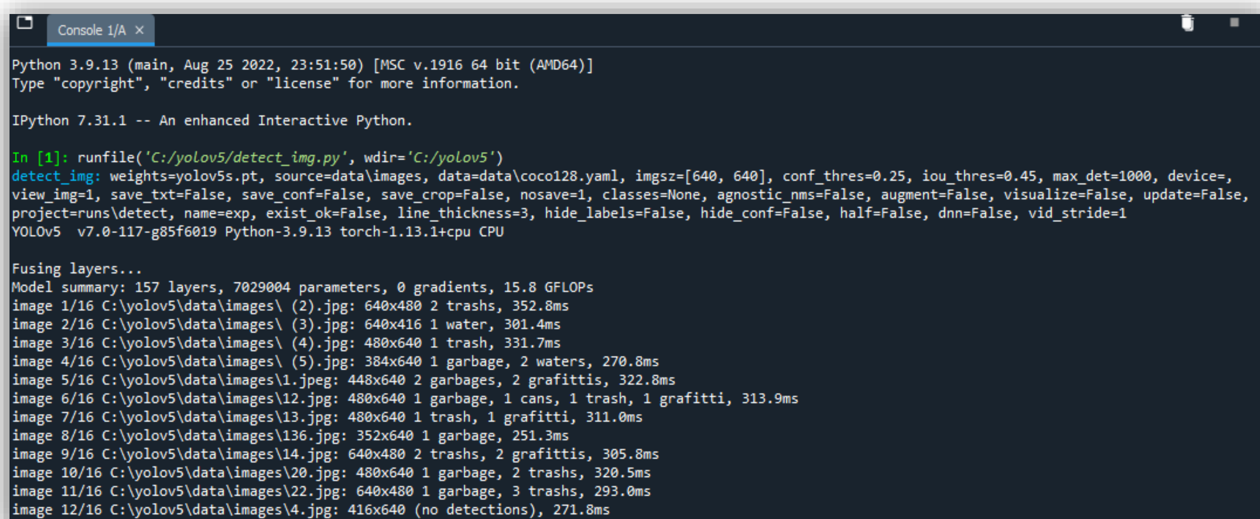
detect_img.txt

```
Anaconda Prompt (anaconda3) - python  detect_img.py

(base) C:\Users\HP>cd\

(base) C:\>cd yolov5

(base) C:\yolov5>python detect_img.py
detect_img: weights=yolov5s.pt, source=data\images, data=data\coco128.yaml, imgsz=[640, 640], conf_thres=0.25, iou_thres
=0.45, max_det=1000, device=, view_img=1, save_txt=False, save_conf=False, save_crop=False, nosave=1, classes=None, agno
stic_nms=False, augment=False, visualize=False, update=False, project=runs\detect, name=exp, exist_ok=False, line_thickn
ess=3, hide_labels=False, hide_conf=False, half=False, dnn=False, vid_stride=1
YOLOv5  v7.0-117-g85f6019 Python-3.9.13 torch-1.13.1+cpu CPU

Fusing layers...
Model summary: 157 layers, 7029004 parameters, 0 gradients, 15.8 GFLOPs
```

Figure 4.4.10.2 The code detection

After selecting some pictures

Figure 4.4.10.3 result



```
Console 1/A

Python 3.9.13 (main, Aug 25 2022, 23:51:50) [MSC v.1916 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 7.31.1 -- An enhanced Interactive Python.

In [1]: runfile('C:/yolov5/detect_img.py', wdir='C:/yolov5')
detect_img: weights=yolov5s.pt, source=data\images, data=data\coco128.yaml, imgsz=[640, 640], conf_thres=0.25, iou_thres=0.45, max_det=1000, device=,
view_img=1, save_txt=False, save_conf=False, save_crop=False, nosave=1, classes=None, agnostic_nms=False, augment=False, visualize=False, update=False,
project=runs\detect, name=exp, exist_ok=False, line_thickness=3, hide_labels=False, hide_conf=False, half=False, dnn=False, vid_stride=1
YOLOv5  v7.0-117-g85f6019 Python-3.9.13 torch-1.13.1+cpu CPU

Fusing layers...
Model summary: 157 layers, 7029004 parameters, 0 gradients, 15.8 GFLOPs
image 1/16 C:\yolov5\data\images\ (2).jpg: 640x480 2 trashs, 352.8ms
image 2/16 C:\yolov5\data\images\ (3).jpg: 640x416 1 water, 301.4ms
image 3/16 C:\yolov5\data\images\ (4).jpg: 480x640 1 trash, 331.7ms
image 4/16 C:\yolov5\data\images\ (5).jpg: 384x640 1 garbage, 2 waters, 270.8ms
image 5/16 C:\yolov5\data\images\1.jpeg: 448x640 2 garbages, 2 grafittis, 322.8ms
image 6/16 C:\yolov5\data\images\12.jpg: 480x640 1 garbage, 1 cans, 1 trash, 1 grafitti, 313.9ms
image 7/16 C:\yolov5\data\images\13.jpg: 480x640 1 trash, 1 grafitti, 311.0ms
image 8/16 C:\yolov5\data\images\136.jpg: 352x640 1 garbage, 251.3ms
image 9/16 C:\yolov5\data\images\14.jpg: 640x480 2 trashs, 2 grafittis, 305.8ms
image 10/16 C:\yolov5\data\images\20.jpg: 480x640 1 garbage, 2 trashs, 320.5ms
image 11/16 C:\yolov5\data\images\22.jpg: 640x480 1 garbage, 3 trashs, 293.0ms
image 12/16 C:\yolov5\data\images\4.jpg: 416x640 (no detections), 271.8ms
```

Figure 4.2.10.4: The result of code detection

UserGUI

```python
import os
import sys
from PyQt5.QtWidgets import QMainWindow, QAction, qApp, QApplication
from PyQt5.QtGui import QIcon
from PyQt5.uic import loadUi
from PyQt5 import uic
from PyQt5 import QtWidgets, QtGui
from PyQt5.QtGui import QPixmap, QImage


import cv2
import numpy as np

from cgitb import enable
import sys
import os

import multiprocessing

############ Model Lib ###############
import os
import platform
import sys
from pathlib import Path
import torch
FILE = Path(__file__).resolve()
ROOT = FILE.parents[0]  # YOLOv5 root directory
ROOT = Path(os.path.relpath(ROOT, Path.cwd()))  # relative
from models.common import DetectMultiBackend
from utils.dataloaders import IMG_FORMATS, VID_FORMATS, LoadImages, LoadScreenshots, LoadStreams
from utils.general import (LOGGER, Profile, check_file, check_img_size, check_imshow, check_requirements,
                           increment_path, non_max_suppression, print_args, scale_boxes, strip_optimizer,
from utils.plots import Annotator, colors, save_one_box
from utils.torch_utils import select_device, smart_inference_mode
```

Figure 4.4.10.3 userGUI

userGUI.py

Result:

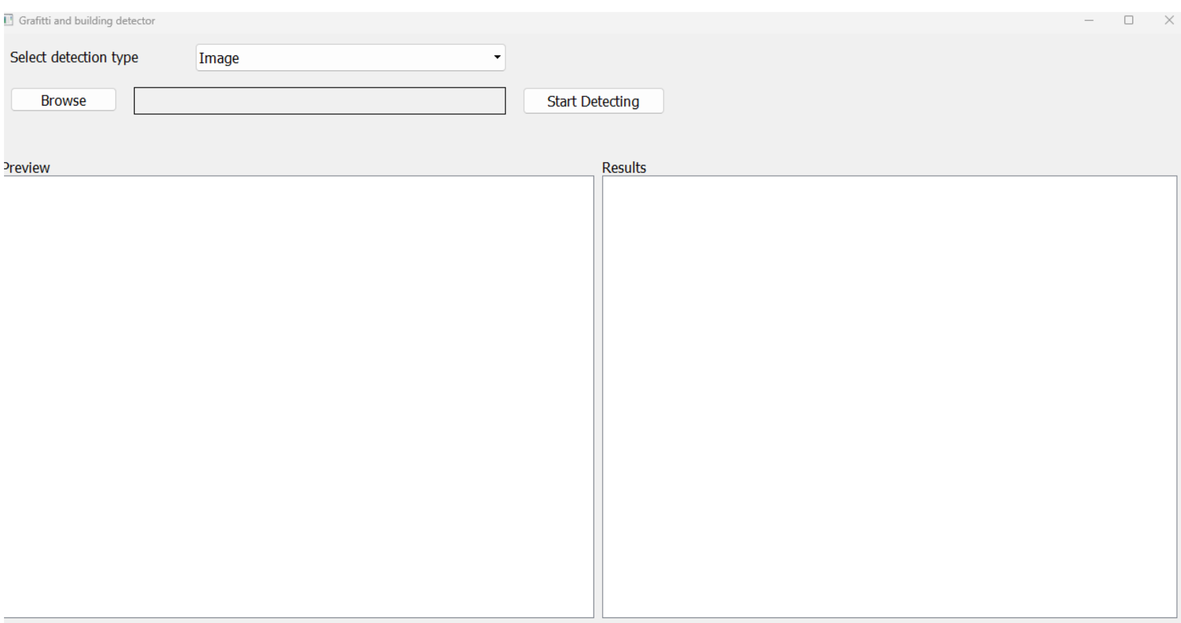


Figure 4.4.10.4: Result

Some pictures:

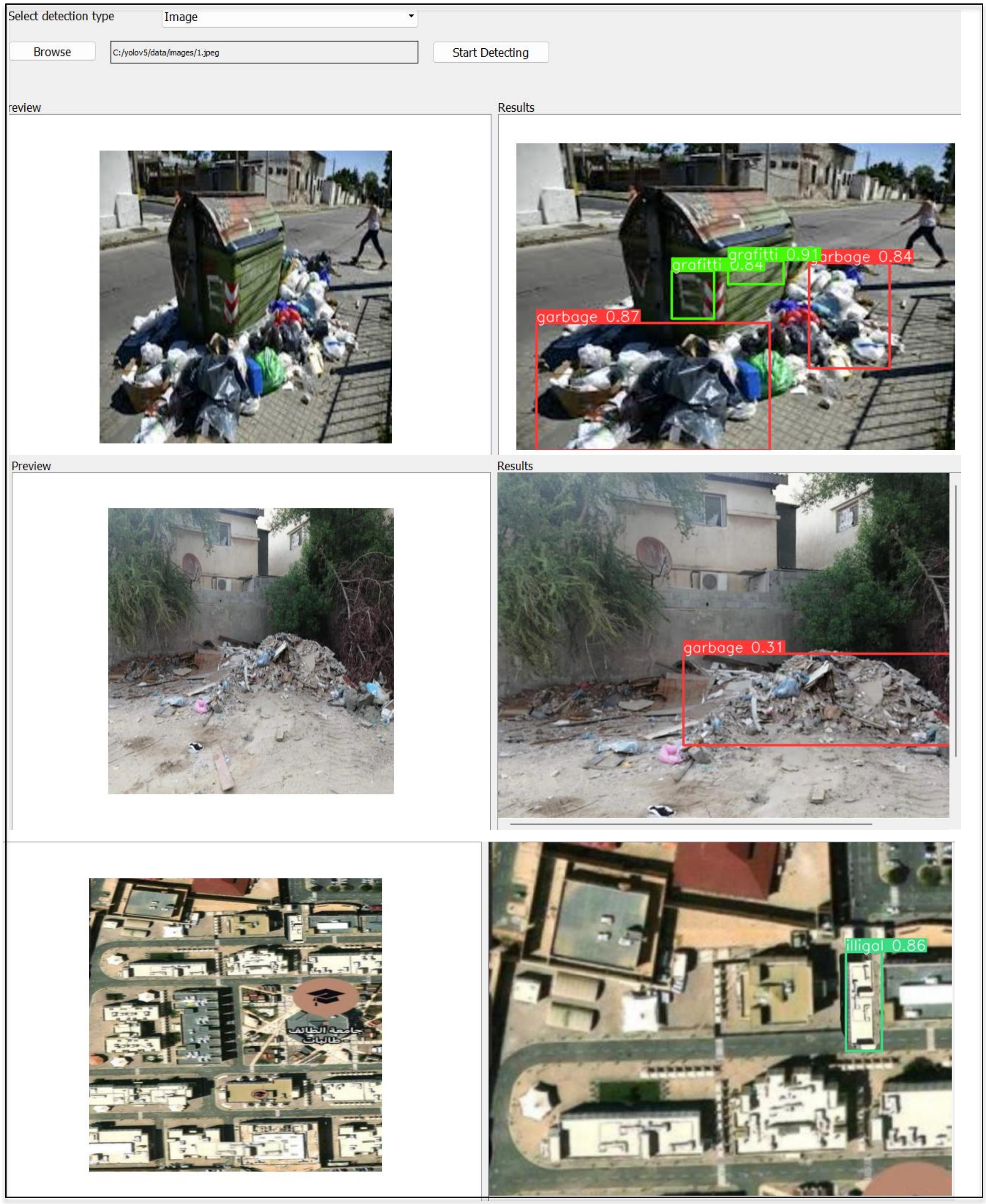


Figure 4.4.10.5: userGUI result

Chapter 5: Practical Implementation and Results

5.1) Sensors and Flying mechanism

5.1.1 Flying mechanism

Firstly, start with rise and fall the Drone. In order for the drone to start rising, the propellers must generate a lifting force that overcomes the force of gravity and the weight of the drone

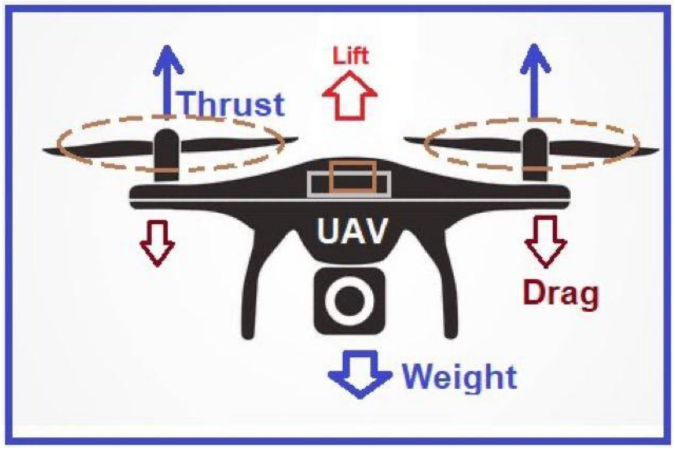


Figure 5.1.1 Flying mechanism

This lifting force results from the rotation of the four fans, which push the air downward, and because each action has a reaction equal in magnitude and opposite in direction, the air compressed downwards in turn pushes the fans upwards, and so the higher the speed of the fans, the greater the lifting force and the greater the height.

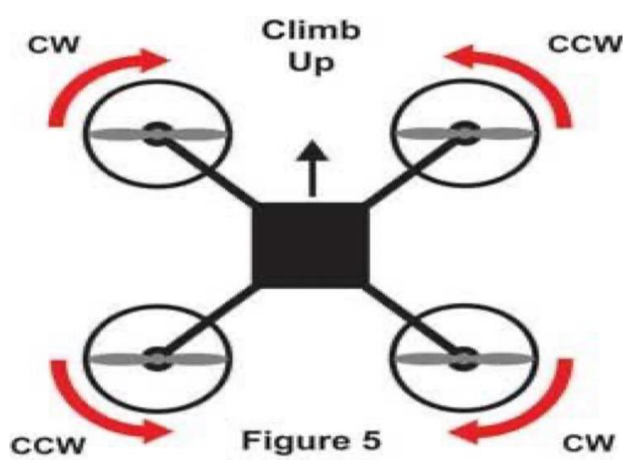


Figure 5.1.2: Flying mechanism

Drone propellers do not all rotate in the same direction, because if they rotate in the same direction, the torque output will push the entire drone body to rotate, which leads to a loss of balance, so to solve this problem, each pair of the four propellers rotates diagonally opposite the direction of the other pair of propellers. Therefore, the direction of the torque of each pair of propellers is in the opposite direction of the other, which results in a torque equal to zero, and thus the plane will get rid of the influence of the torque and it rises steadily without losing balance.

Secondly Drone rotation around its axis (YAW) when the drone rises in the air and the controller wants to change its direction to the right or left to direct it towards a specific direction (i.e. its rotation in its position), what really happens is to increase the speed of two of the diagonal motors (fans) and reduce the speed of the other two

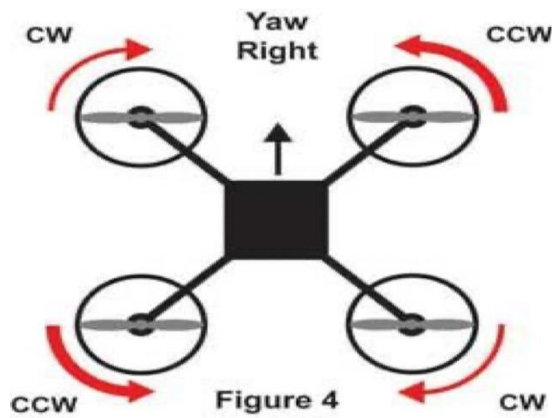


Figure 5.1.3.1: Flying mechanism

This results in an increase in the torque of a pair of diagonal propellers and a decrease in the torque of the second pair of propellers, which causes the drone to rotate in the opposite direction of the propellers whose rotation speed has been increased.

Go forward, backward, right, or left (pitch/roll) to move the drone forward, the controller reduces the speed of the front motors and increases the speed of the rear motors, which results in tilting the drone forward, resulting in an imbalance in the drone.

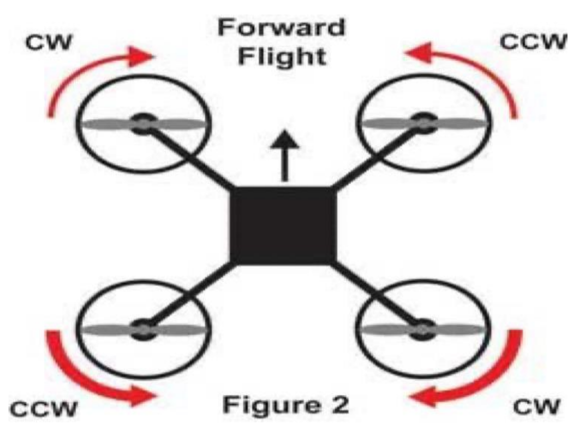


Figure 5.1.3.2: Flying mechanism

When the controller tries to fix its height, the drone automatically rushes forward, and vice versa, to move it backwards, and the same idea is the direction towards it, right or left. Let's imagine that the user was able to control the speed of the propellers and balance the drone, and then it was exposed to winds that caused a defect in balancing it. In this case, the user will need to control the speed of the propellers to balance the drone in less than a second, otherwise it will lose balance and fall. It is very difficult for the user to be able to balance and direct the drone by controlling the speed of the propellers directly. For this reason, we need a flight controller to calculate the drone's inclination angle, speed, and height so that it can balance and direct the drone according to the commands received from the user.

5.2.2) Sensors used by the flight controller:

Flight Controller it is a microcontroller equipped with sensors to calculate the angle, direction, speed and height of the drone so that it can control the speed of the motors to balance the drone and direct it according to the commands received from the user.

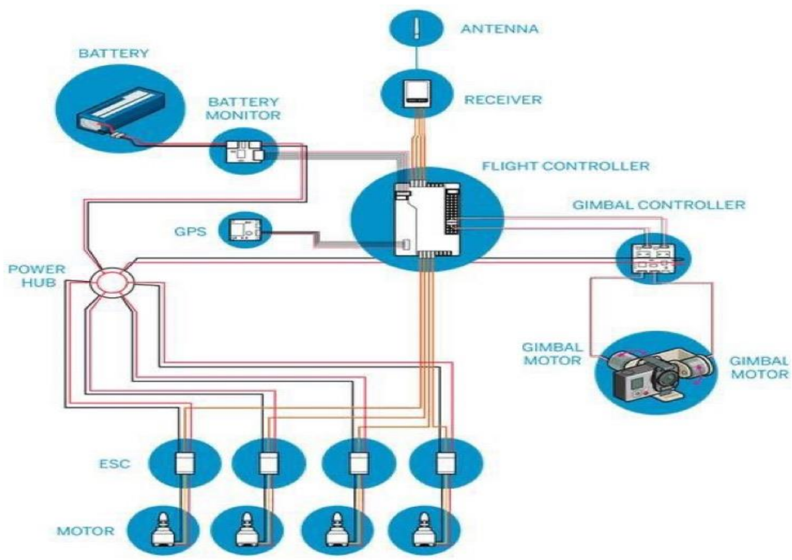


Figure 5.2.2: Sensors used by the flight

The most important sensors used by the flight controller:

1-Inertial measurement unit (IMU):

It is an electronic piece that measures and determines the forces acting on the body, its angular acceleration, and its direction. The IMU is made up of a gyroscope and an accelerometer that can both generate linear acceleration signals on three axes of space and angular velocity data on the same three axes. Common uses of IMUs include detecting motion in consumer electronics like mobile phones and video game controllers, identifying direction in GPS systems, and tracking head motions in augmented reality and virtual reality systems.

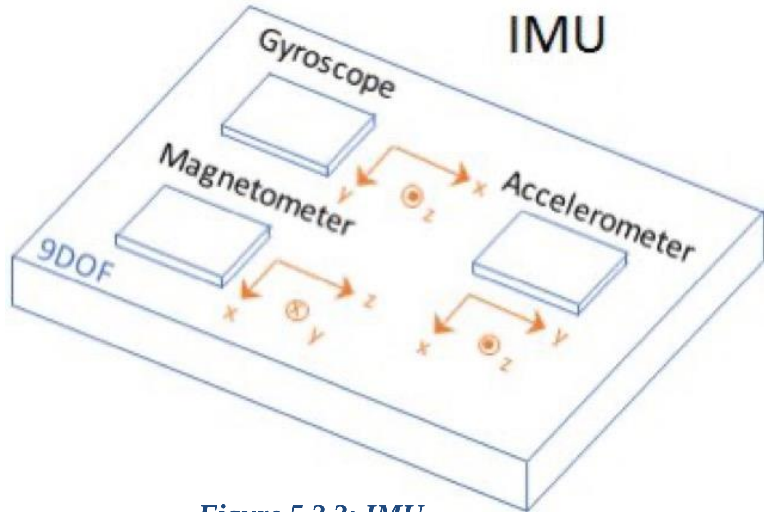


Figure 5.2.3: IMU

The IMU (Impedance Measurement Unit) consists of several sensors:

- Accelerometer to calculate the acceleration of the body of the drone.

- A gyroscope sensor to calculate the amount of change in the angle of the drone's body.

- The digital compass or (magnetometer) to determine the direction of the body of the drone.

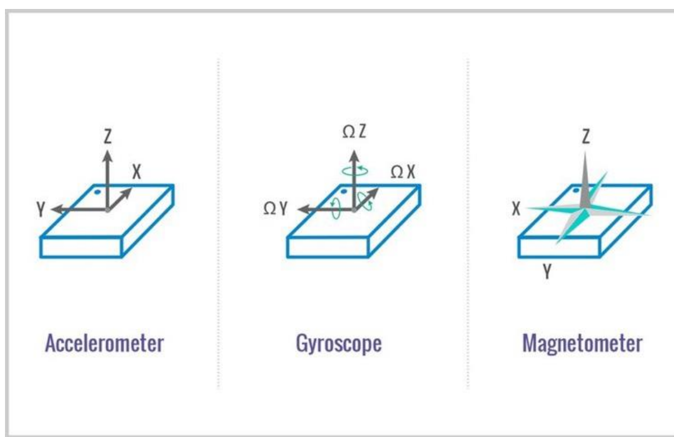


Figure 5.2.4: Sensors

2-Barometer: to measure atmospheric pressure so that the controller can calculate the height. A barometer, often known as a barometric pressure gauge, is a scientific device used to measure atmospheric pressure. The air layers that surround Earth are known as the atmosphere. When gravity drags that air to Earth, it exerts pressure on everything it touches. This pressure is measured using barometers.

3-Satellite Navigation System (GPS): to locate the drone. Is a network of ground stations and satellite control stations used for monitoring and control, as well as a constellation of satellites that transmit navigation signals.

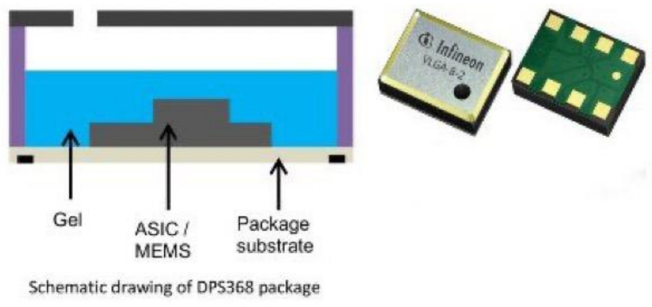


Figure 5.2.5: GPS

5.3.1) Flight controller function

The function of the flight controller is divided into three main parts

1- Sensing (sensor):

A microcontroller uses an IMU to calculate acceleration and angular velocity and makes a barometric pressure and altitude sensor, spearing these sensors to a lot of noise (Nosie signal) for example: the vibration resulting from the rotation of the motors causes an error in measuring acceleration, angular velocity, and direction, and thus the stability of the drone in the air will be affected or lead to its fall or the compass (magnetometer) is exposed to interference with the electromagnetic field resulting from the passage of electric current in the wires. This interference causes an error in determining the direction. The flight controller receives data from these sensors about the aircraft, including its height, orientation, and speed. Typical sensors include a barometer for height, an inertial measurement unit (IMU) for angular speed and acceleration, and distance sensors for obstacle detection.

2- Communication:

The flight controller receives commands from the remote control via radio waves

- The flight controller can send drone status data to the user, such as battery status, altitude, and direction. Aviation communication is the method by which pilots and ground personnel exchange information between themselves and other aircraft. The smooth operation of aircraft movement both on the ground and in the air depends heavily on aviation communication.

3- Control (Controlling):
The flight controller uses the data it gets from the sensors to calculate the required speed for each engine
Then it sends the required speed to the speed controllers (ESC) to move the motors at the required speed.

5.4.1) Algorithms of flight control:

The flight controls are the most crucial algorithms.

1-PID Controller: Also known as a proportional + integrative + differential controller, this device is in charge of resolving errors brought on by discrepancies between desired and measured values.

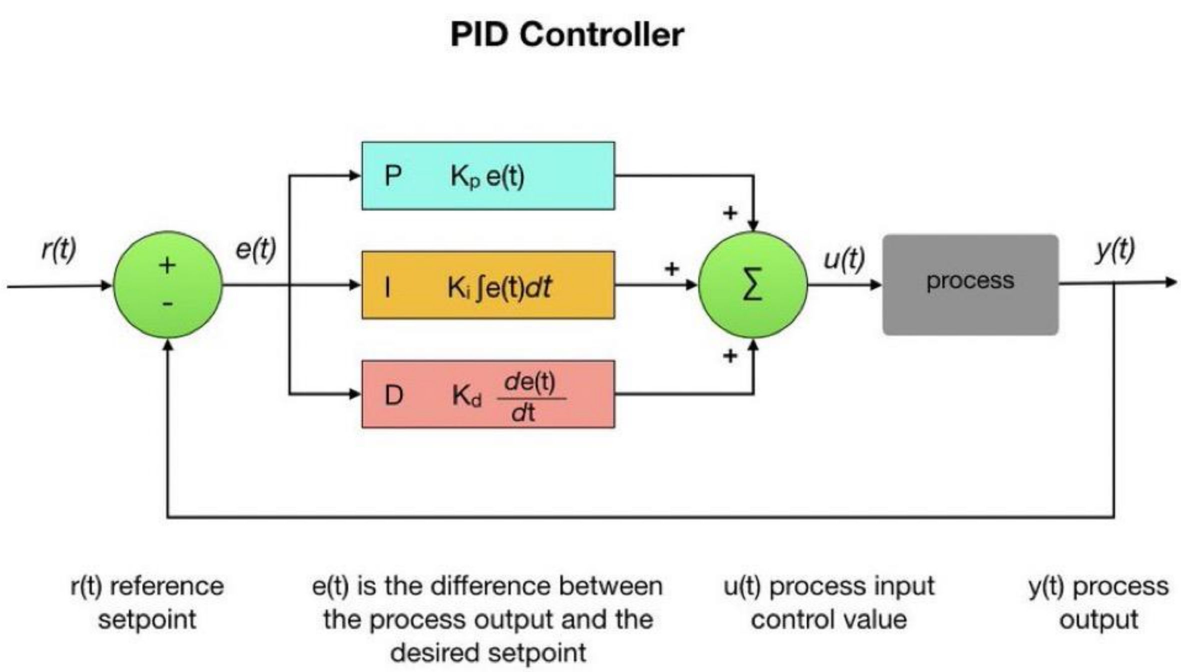


Figure 5.4.1.1: PID controller

"Set-point" "r(t)" denotes the intended rotational rate. The "error" "e(t)" represents the variance between the set-point (the rate at which we want the drone to rotate) and the measurement from the gyro sensor (how fast the drone is actually rotating).

The main objective of a PID controller in a drone is to repair the "mistake" by modifying motor speeds. In order to reduce error, the control loop continuously analyzes sensor data and calculates motor speeds.

A key component of the control system is the PID algorithm. A PID controller uses the terms proportional ("P"), integral ("I"), and derivative ("D") (D).

1. The existing inaccuracy is related to P (proportional). In mathematics, it is proportional to the error, meaning that the higher the error, the harder it pushes.

2. The derivative D forecasts upcoming mistake. In mathematic terms, it is the derivative of the error since it takes into account how quickly the set-point is reached and counteracts P to reduce overshoot when approaching the target.

3. I (integral) builds up previous mistakes. By modifying motor speeds to oppose it, it handles external pressures that develop over time, such as a drone drifting off set-point owing to wind or an off-centered weight - in math terms, it is the integral of the error.

Your drone will track stick movements accurately when PID is calibrated properly, giving you a direct, responsive feel free of wobbles or oscillations.

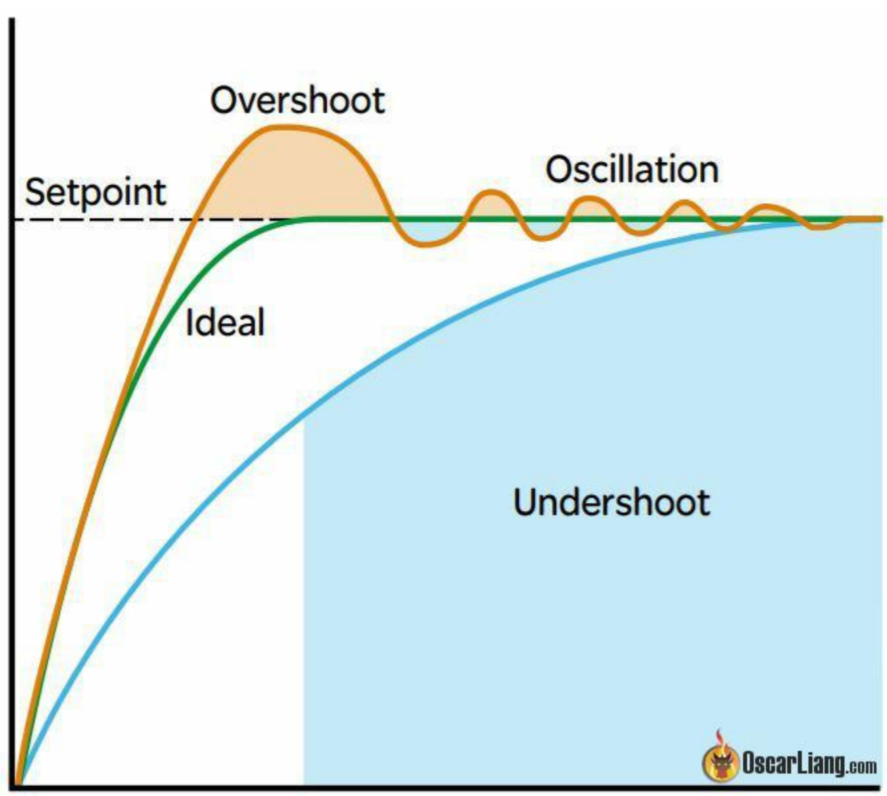


Figure 5.4.1.2: PID

The Effect of PID Terms:

A quadcopter's flight behavior is affected by adjusting PID gains. Although it is not necessary to comprehend PID's inner workings, it is important to understand how modifying these gains effect your drone's performance.

• P Gain:

P gain controls how hard the flight controller attempts to remedy faults. As a responsiveness setting, think of P gain. A quick reaction is produced by a large P gain, giving the impression that your rates have increased. The quadcopter tends to overcorrect when P gain is too high, which causes abrupt bounce-backs during flips and rolls. High P gain levels go so far as to produce oscillations. In contrast, the quadcopter seems sloppy and slow to react when P gain is too low.

• I Gain:

The amount of effort put forward by the flight controller to keep the drone's attitude despite external factors like wind and an off-centered CG (center of gravity). Consider I gain as a stiffness parameter that controls how effectively the quadcopter maintains its attitude. Although higher I gain stiffens the quad's entry into a move, it enhances tracking set points during sweeping turns. But at a considerably slower (lower frequency) rate, high I gain can also result in bounce backs and oscillations. The quad will shake and "nose dive" after quick throttle adjustments if I gain is too low. In windy situations, extremely low I gain leads to poor angle-holding, which makes the drone feel like it is drifting and necessitates regular pilot input for corrections.

• D Gain

P gain overshoots are decreased by D gain acting as a damper. Similar to how a shock absorber reduces suspension bounce, D gain reduces oscillations brought on by excessive P gain and prop wash oscillations. When D gain is too low, the quadcopter will have greater prop wash oscillations during vertical descents and will bounce back significantly more after flips and rolls. These problems can be reduced by increasing D gain; however, too much D gain might increase noise and vibrations inside the quadcopter, leading to trilling oscillations and motor overheating.

2-Sensor fusion:

This technique boosts the precision of calculations by integrating the data from two or more sensors.

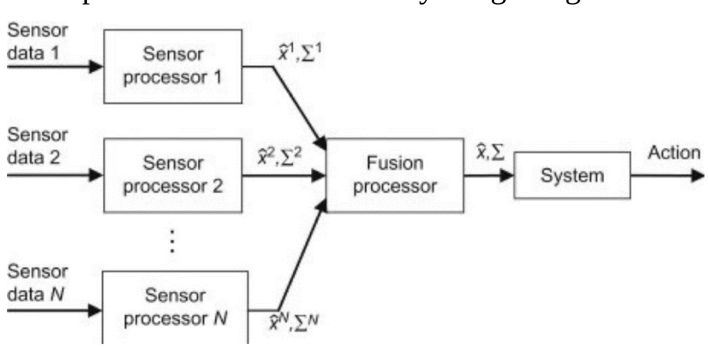


Figure 5.4.1.3: Sensor fusion

For instance, by integrating various data sources, such as video cameras and WiFi localization signals, one may be able to get a more precise location estimate of an indoor object. In this context, the term "uncertainty reduction" can either relate to the outcome of an emerging view, like stereoscopic vision, or it might indicate "more accurate, more complete, or more dependable" (calculation of depth information by combining two-dimensional images from two cameras at slightly different viewpoints).

It is not required that the data sources for a fusion process come from identical sensors. There are three types of fusion: direct, indirect, and fusion of the outputs of the first two. Whereas indirect fusion relies on data sources including a priori environmental knowledge and human input, direct fusion combines sensor data from a variety of heterogeneous or homogeneous sensors, soft sensors, and sensor data historical values.

3-Extended Kalman Filter (EKF):

The drone is less prone to error since it incorporates all available data and discards any that include errors. It also evaluates the drone's condition by determining its location and speed.

Drones have senses. Drones can compute their current position and speed thanks to these sensors, which enable them gather information about their surroundings. It is possible to combine several sensors to improve the data. Unfortunately, because of the noise that the sensors produce, the data that is collected from them may be erroneous. Due to the possibility of each sensor reading providing data that is inconsistent with one another, this causes the drone to go crazy. If so, how does the drone determine its location and its direction of travel? The best estimate of the drone's current position and velocity is produced using a kalman filter (KF), which uses the data provided by the sensor in combination with an analysis of the probability distribution of each sensor. We can utilize more reliable varieties of KF that produce better estimates to enhance the filter. The extended kalman filter (EKF), which takes into account nonlinear data, is a more reliable variety.

why do drones require sensors?

Drones feature sensors that allow them to understand their current situation, learn more about their environment, and decide how to respond. These sensors will serve as the drone's eyes and hearing during the entire flight because we are dealing with an autonomous drone. The drone has two properties thanks to its sensor. situational awareness, which enables the drone to identify and track nearby objects, and self-awareness by providing the drone with information about its location and position in time. Four different types of sensors will be used by the drone in our project:

- Accelerometer: It records the drone's linear movement. It detected any changes in the drone's velocity, including the gravity-induced steady downward acceleration.
- Gyroscope: It measures the angle of the drone's movements. This sensor aids in determining the angle of tilt, rate of rotation, and angular velocity of the drone.
- Barometer: It gauges the pressure of the air around the drone. By calibrating the sensor in relation to sea level, the drone can detect its height because air pressure decreases correspondingly with altitude.

• Camera: The drone will be more aware of its surroundings with the aid of this sensor. The other sensors previously stated have so far assisted the drone's self-awareness. The situational awareness of the drone will only be assessed using this sensor.

Used when:

• Measurements are available from a variety of sensors, although they may be noisy.

• The variable of interest can only be monitored indirectly.

5.2) Experiments with Mission Planner:

We used the software Mission Planner for the aim of the drone's calibration. [33]

Figure 5.2.1.1: Mission Planner interface

5.2.2) Calibration of Accelerometer:

The frame layout and copter are selected as shown in figure below

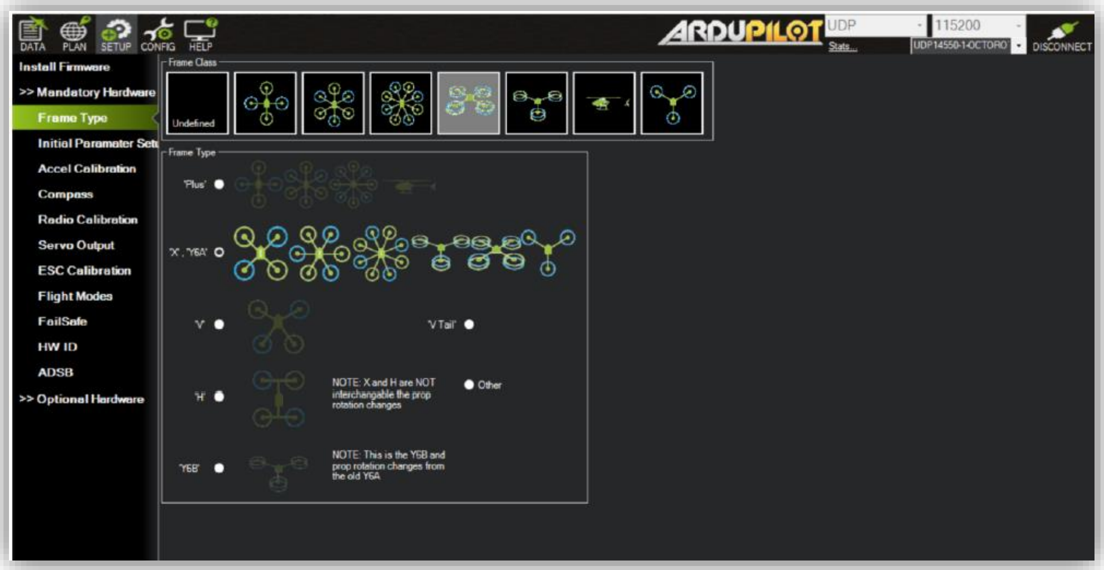


Figure 5.2.2.1: Selection of frame layout

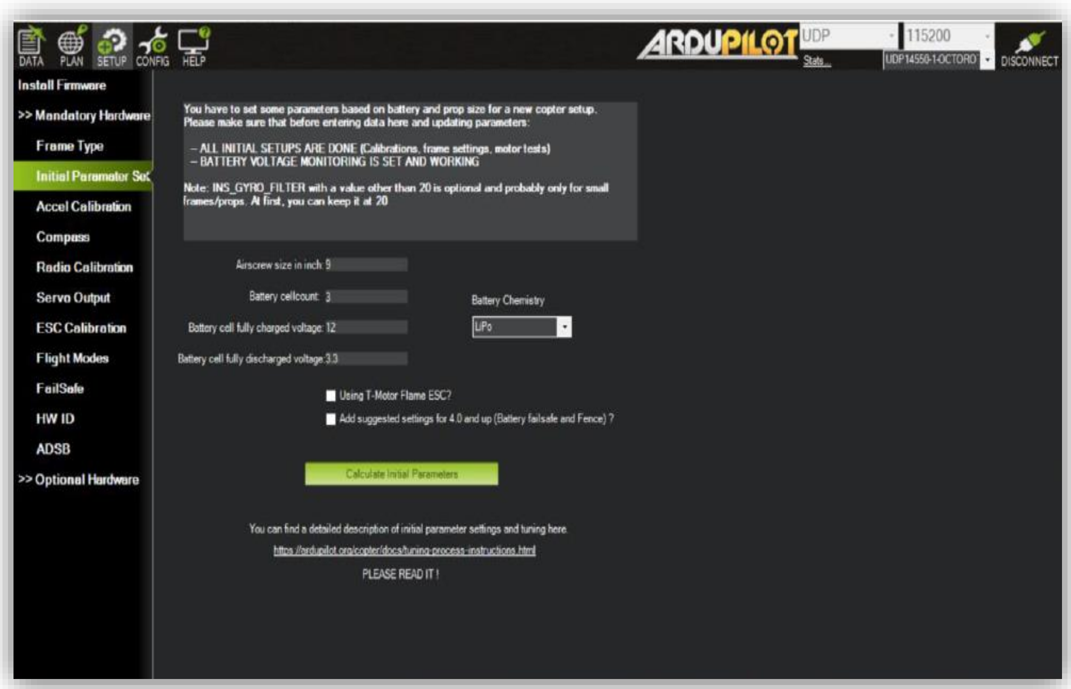


Figure 5.2.2.2: Prameter set

Afterwards, ***calibrate accel*** was chosen and we followed the instructions

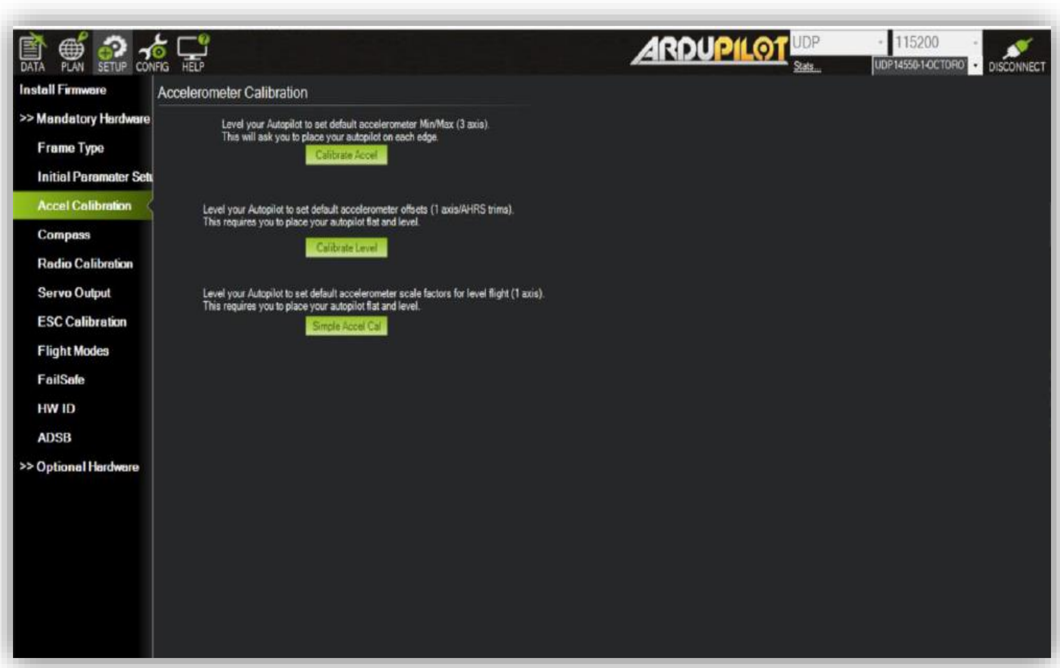


Figure 5.2.2.3: Accelerometer calibration

The software then asked us to place the drone in the correct calibration position. These positions are: on the right side, left side, nose down, nose up, and back. We applied the right-side calibration as shown below.

Figure 5.2.2.4: Calibration right side

Figure 5.2.2.5: Calibration left side

Figure 5.2.2.6: Calibration on nose down

Figure 5.2.2.7: Calibration on nose up

Figure 5.2.2.8: Calibration on back-side

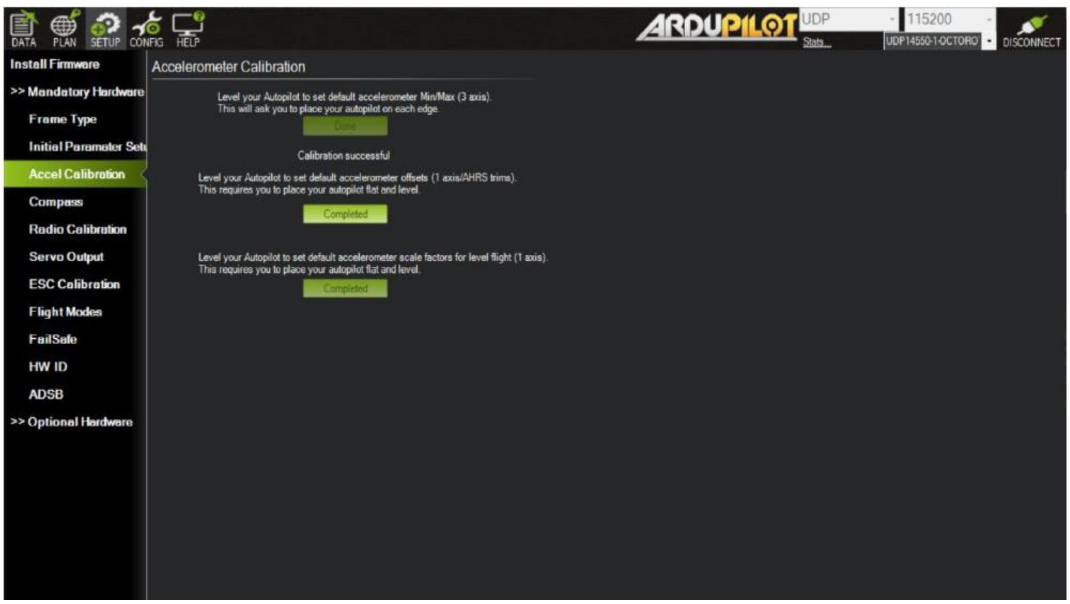


Figure 5.2.2.9: Calibration Successfully

As soon as we completed the calibration process, Mission Planner display "Calibration Successful".

5.2.3) Calibration of Compass:

The drone was held in the air, and it was rotated to each side (front, back, left, right, top and bottom), and pointed down towards the earth for a few seconds in turn.

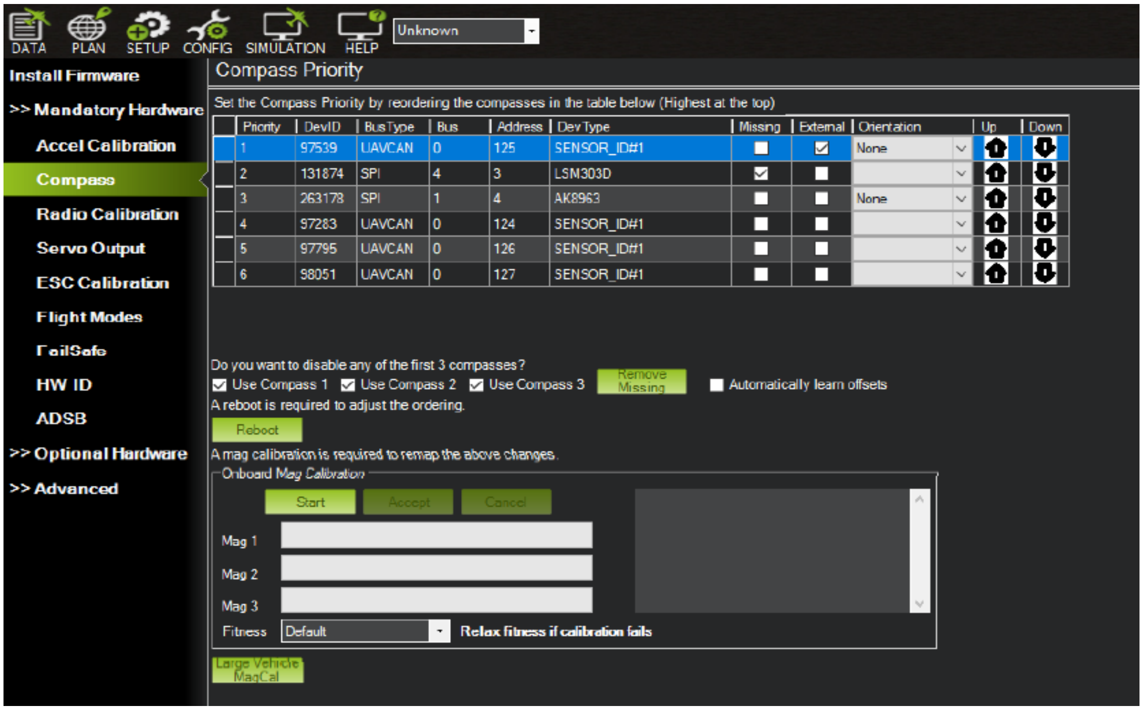


Figure 5.2.3: Compass Setting

5.2.4) Radio Control Calibration:

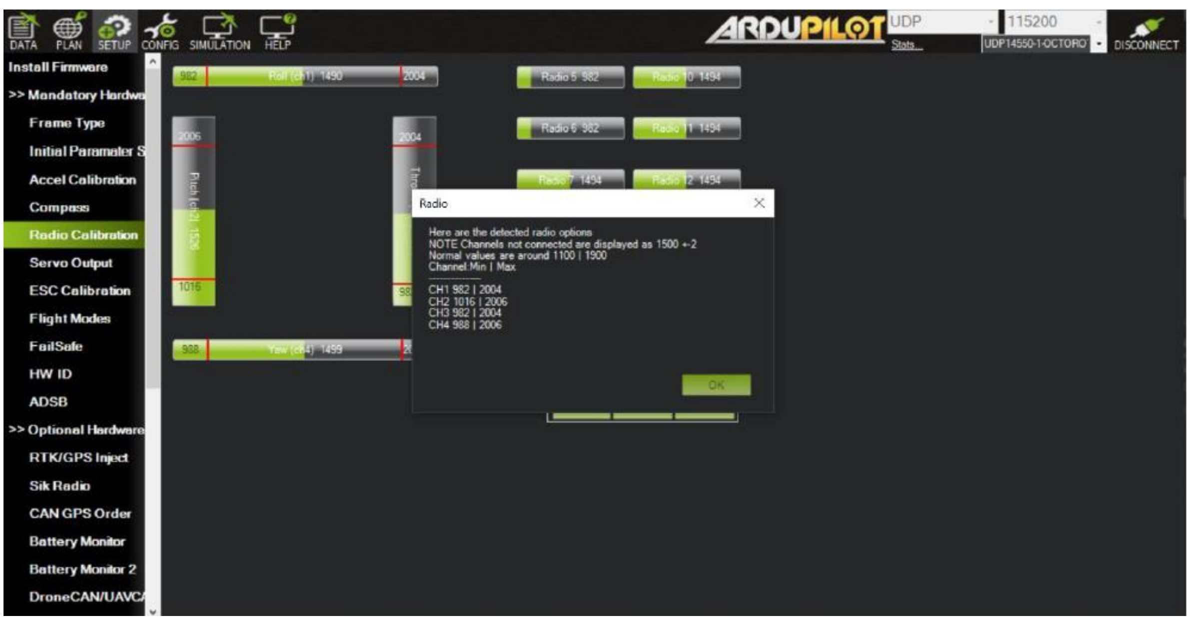


Figure 5.2.4: Radio Control Calibration

5.2.5) ESCs Calibration:

Figure 5.2.5: ESCs Calibration

Transmitter was turned on; the throttle stick was set to the maximum and the Lipo battery was connected successfully. The LEDs red, blue and yellow of the autopilot were lit up all in a periodic pattern, which deducts ESC calibration mode was ready to go in the next time we plug it in. If transmitter throttle stick is high, battery's disconnection and reconnection must be achieved. The emitted musical tone by the ESCs, help us in estimating the battery's cell by watching the regular peeps number. For instance, 3 beeps for 3 seconds, as well, extra 2 beeps will mean the maximum throttle has been captured. After waiting, transmitter's throttle stick should be pulled down to its minimal position. If the ESCs emitted a long tone, this indicates that the minimal throttle was captured, and the calibration was completed.

At the end, the throttle should be set to its minimal position and the batteries should be disconnected to exit the ESC-calibration mode.

5.3) OpenHD:

OpenHD is a suite of software designed for long-range video transmission, telemetry, and RC control. While we originally designed it with hobbyist drones in mind, it can be adapted to a wide range of other applications as well.

5.3.1 What OpenHD Can Do:

Can transmit:

- High-definition video (with multiple cameras)
- Two-way UAV telemetry
- Audio
- RC control signals

5.3.2 How OpenHD Works:

We use off-the-shelf Wi-Fi adapters that are available for purchase online. However, we don't use standard Wi-Fi, which isn't well-suited for low-latency or long-distance transmissions. Instead, we configure the Wi-Fi adapter to operate like a simple broadcast, which is more similar to analog video transmission hardware.

This configuration allows us to transmit high-quality video and other signals over long distances with low latency.

Figure 5.3.1: OpenHD

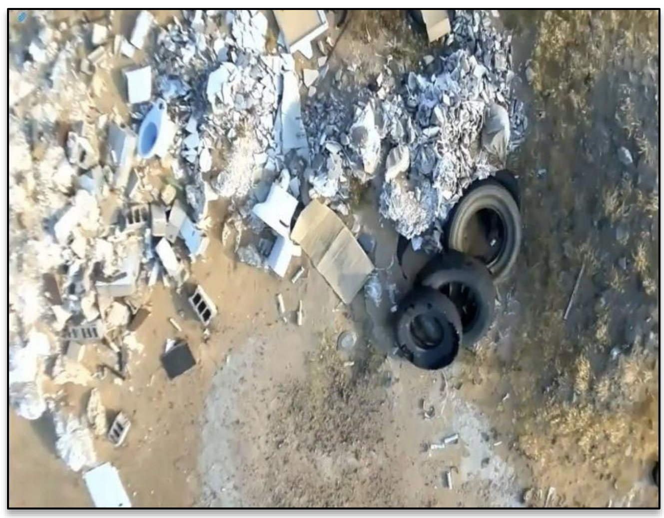


Figure 5.3.2: OpenHD

5.4) Final prototype of drone:

Figure 5.4: Final Drone

Chapter 6: Conclusions and Future Work

6.1) Conclusions

In this research, we designed and developed a system for features where we may use a fire detection mechanism and rainwater harvesting places to lessen the visual deformation inherent in slums and because these random areas have poor discharge. Additionally, we may organize neighborhoods and lessen street encroachment. Solving these issues and lowering uncivilized violations are crucial

6.2) Recommendations and Future Work

A Drone has the ability to perform many tasks, where we can reduce the visual deformation inherent in slums and rainwater harvesting areas because these random areas have poor discharge. We can also reduce street encroachment and organize neighborhoods. So, it is possible to design a robot in the future connected to the drone so that the drone will photograph excavations in the streets and lampposts, visual deformation like garbage and graffiti, etc. and send it to the robot, and the robot will fix it and work on it.

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Appendix A General Remarks

Materials that may be of interest to some readers but not essential to be in the body of the report go to the appendices. This includes:

- Long tables that contain materials not essential
- Locally developed research aid forms, questionnaires
- Analysis data and other materials
- Listing of computer programs

- Students must discuss with their Supervisor the need for appendices.
- Appendices must be designated with letter (Appendix A, Appendix B, etc) each starting on a new page, with a title.
- Each appendix must be listed in the Table of contents.
- **Figures and Tables**

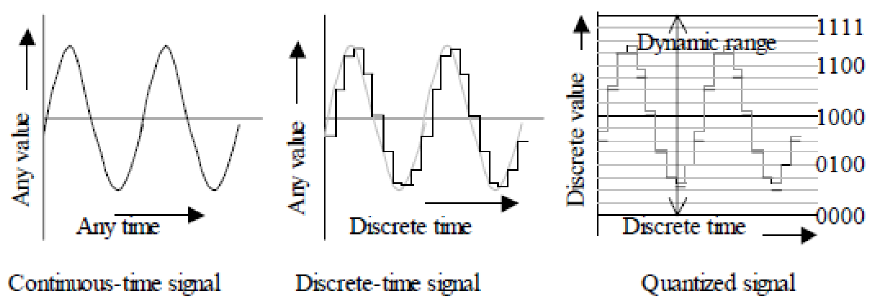

Fig. 1.3 Principle of operation of the analog to digital conversion [2]

Figure caption (title)

Reference

- Technical reports only contain Figures and Tables.
- Refer to graphs as figures, small code segments as figures, etc.

Table 2.1 Conditions of pins of the parallel port

Table caption

Pin #	Pin name	Connection
1	$\overline{C0}$	Inverted
14	$\overline{C1}$	Inverted
16	$C2$	Non Inverted
17	$\overline{C3}$	GND
15	$S3$	Non Inverted
13	$S4$	Non Inverted
12	$S5$	Non Inverted
10	$S6$	Non Inverted
11	$\overline{S7}$	Inverted

- Tables and figures must support the text.
- References to figures and tables, use for example:
  - Fig. 2.2 (or figure 2.2)
  - Table 3.1
- Usually captions for:
  - Tables go above the table.
  - Figures go below the figure.
- Do not use shortened forms such as “don’t” for “do not”.
- Try to write in “third person”, first person “I” can be used when stating an opinion. Example “A test was run….” NOT “I ran the test…..”.
- Past tense is used for explaining procedures.
- Present tense is used for generalization and for stating what the results show.
- Whenever possible use lists.
- Be specific; avoid vague statements such as “the results were very promising”.
- Write body of report first before abstract.
- Use lower case Roman numerals (i, ii, iii etc.) for preliminary pages, and Arabic numerals (1,2,3, etc.) for the body of the report, starting with page 1 for the introduction.